Adjoint Gradients for Torque-Ripple Optimization of a Nonlinear Motor

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Abstract

A design gradient pays off only if it shortens the path to a validated design. We reconstruct a published nonlinear permanent-magnet motor in our GEODE finite-element code and minimize its torque ripple over 8 or 18 shape parameters. The optimizers query a coarse simulation; each new best design is rescored on a finer validation simulation no optimizer sees, and only validated scores count. Cost is whole-run wall time on one single-threaded core per run. In a preregistered, budget-matched study, SLSQP with adjoint gradients beat the derivative-free COBYQA at the same coarse resolution in all 45 of the 50 paired comparisons where both succeeded, by geometric means of 3.10 and 7.79 at the primary targets, 90% and 99% reductions of the ripple objective with 8 and 18 parameters. Two post-hoc ablations, under protocols frozen beforehand, kept SLSQP but used forward differences, splitting that advantage into a gradient factor and an optimizer factor. The adjoint reached the primary target 1.71 and 2.70 times sooner, from every start, than forward differences with warm-started solves, and 2.47 and 4.11 times sooner than cold-started ones; COBYQA against finite-difference SLSQP contributes the remaining 1.2 to 1.8 with 8 parameters and 1.9 to 2.9 with 18. The adjoint cuts the gradient's share of the time to target from 45–77% to about 5%, so validation and coarse-simulation evaluations bound the end-to-end gain. Every target crossing selected for final checks, 25 of the main study's 111 and 10 in each ablation, kept its target on a finer mesh.

1 Introduction

Whether to pay for derivatives is a practical design decision. An adjoint gives the sensitivities of a few outputs to many parameters for a cost that does not grow with the number of parameters, whereas a finite-difference gradient needs one extra solve per parameter (Giles and Pierce, 2000; Martins and Hwang, 2013). That does not settle whether following adjoint gradients reaches a good design sooner than a derivative-free search or a cheaper gradient, or whether the design survives a better model: the adjoint differentiates the chosen discrete model and excitation policy. In aerodynamic shape optimization an adjoint gradient-based method has needed far fewer function evaluations than a genetic algorithm (Zingg et al., 2008), so the direction of an advantage is expected. Its size for a nonlinear motor, measured on a validated target with failures retained, is the question here.

We answer it in two steps. A preregistered, budget-matched study compares five methods in two nested design spaces of a published nonlinear permanent-magnet motor (Wiesheu et al.,

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2024), judges every candidate on one validation simulation, finer than any coarse simulation an optimizer queries, and charges whole-run wall time with nothing subtracted (Section 5). There SLSQP with adjoint gradients beats COBYQA, but the contrast changes the search strategy and the gradient together. Two post-hoc ablations therefore keep SLSQP and replace the adjoint by forward differences (Section 6), so that the main contrast factors into a gradient factor (finite-difference SLSQP against adjoint SLSQP) and an optimizer factor (COBYQA against finite-difference SLSQP).

Table 5 gives the main result. On one core per run, with the finite-difference stencil (one perturbed coarse solve per parameter) solved serially, the gradient factor is 1.71–2.47 with eight parameters and 2.68–4.10 with 18, and the optimizer factor is 1.22–1.76 and 1.91–2.92. The range in each factor is the gap between the two finite-difference baselines: stencil solves warm-started from the base point's solution (FD-warm) or started from zero (FD-cold). The gradient factor is set by the cost of the gradient, since every SLSQP variant, adjoint or finite-difference, crosses each primary target after the same number of validated designs, and it is bounded by the time all of them spend on validation and coarse forwards (Fig. 7).

Two checks make these measurements interpretable. The ripple objective depends on the rotor–stator interface model: truncating harmonic coupling at the reference's order 45 changes loaded ripple by 7–9% and cogging by 59% on the finest mesh, so the study validates at order 93 (Section 3). The design gradients pass finite-difference and Taylor checks, yet verified ripple sensitivities differ between interface/estimator models, and the excitation rule reverses mean-torque sensitivities (Section 4). The numerical methods are established; the contribution is the measured time-to-target evidence for adjoint gradients, separated from the choice of optimizer, with the checks that show what the gradients represent. Section 7 states what transfers.

2 Related work

Wiesheu et al. (2024) combine spline-based rotor and stator shape optimization with engineering parameters for a nonlinear permanent-magnet synchronous motor, and their related study combines parameter and shape optimization with isogeometric analysis (Wiesheu et al., 2025). Their public implementation and saved outputs provide the reproduction target. We inherit the initial geometry, material table and excitation conventions and replace the field discretization; the comparison is between numerical models, not with a measured motor.

Adjoint shape sensitivity for nonlinear magnetostatics is long established. Park et al. (1992) applied the adjoint variable method to the finite-element equations of a saturated two-dimensional magnet design, and El Bechari et al. (2022) detail a discrete adjoint for linear and nonlinear magnetostatics with geometric and physical variables, verified against finite differences. Gangl et al. (2015) derive rotor shape sensitivities with a shape-Lagrangian formulation to smooth the rotation pattern of a motor; Kuci et al. (2017) derive shape sensitivities from the Lie derivative and apply them to an interior permanent-magnet machine (Kuci et al., 2016). Scotzniovsky et al. (2025) combine free-form deformation, differentiable mesh warping and adjoint derivatives, and Merkel et al. (2021) use isogeometric shape derivatives with harmonic rotor–stator coupling. Martins and Hwang (2013) review finite differences, complex steps, algorithmic differentiation and direct and adjoint methods in one framework, Giles and Pierce (2000) introduce the adjoint approach to design, Nikolova et al. (2004) review adjoint-variable sensitivities for electromagnetic structure design, and Peter and Dwight (2010) and Kenway et al. (2019) survey sensitivity-analysis approaches and adjoint practice for aerodynamic optimization. These works establish why an adjoint gradient is cheaper than a finite-difference one; Section 6 measures what that saving is worth in time to a validated target. The accuracy of a finite-difference gradient also depends on its step relative to the function's computational noise, which Moré and Wild (2011) show how to estimate and Moré

and Wild (2012) use to choose a near-optimal step; our finite-difference baselines instead use one step fixed in advance. For iteratively solved problems the perturbed solves' convergence error enters as well: Haftka (1985) gives a modified procedure that improves such derivatives and Eyi (2011) chooses the step from an estimate of that error, whereas our warm-started baseline keeps the frozen tolerance and reports the resulting error. The direct method, one linearized solve per parameter (Adelman and Haftka, 1986), and the semi-analytic method, which can be inaccurate for shape variables (Barthelemy and Haftka, 1990), lie between the two in cost; neither was tested.

Interface order already matters for torque: De Gersem and Weiland (2004) formulate harmonic weighting functions at the sliding interface, Bontinck et al. (2018) use harmonic interface constraints between isogeometric subdomains and analyze their stability, and Egger et al. (2022) derive torque from a harmonic mortar formulation and vary the transmitted order in a linear cogging experiment. We quantify its effect on loaded nonlinear ripple and design sensitivities, using coupling torque and the gap-averaged stress of Arkkio (1987); Howe and Zhu (1992) and Arkkio and Hannukainen (2012) study discretization effects on torque.

Optimizer benchmarks in machine design compare derivative-free methods: Duan and Ionel (2013) compare response-surface and differential-evolution search on a five-variable permanent-magnet motor, and Bramerdorfer et al. (2018) survey optimization practice for electrical machines, and Lei et al. (2017) review their design-optimization methods. Ion et al. (2018) compare deterministic gradient-based and stochastic search for electric devices. Gangl et al. (2022) report minutes for gradient-based free-form shape optimization of a synchronous reluctance machine against hours for a stochastic parametric optimization that found similar shapes, a comparison that also changes the parametrization and the solver. Zingg et al. (2008) found that a genetic algorithm needed 5 to 200 times as many function evaluations as an adjoint method in aerodynamic shape optimization. On an eight-variable wing twist problem, Yu et al. (2018) found gradient-free optimizers 4 times (NOMAD) to over 100 times (NSGA2, which converged to a slightly worse design) as expensive as adjoint-driven ones, the counterpart of our eight-parameter SLSQP-against-COBYQA ratios of 3.10–3.25. They do not compare adjoint with finite-difference gradients, which our ablation does, but state its premise: finite-difference cost grows with the number of variables and adjoint cost does not. Our time-to-target measure follows the performance profiles of Dolan and Moré (2002) and the data profiles of Moré and Wild (2009), with a wall-time budget and censored targets.

3 Motor model

3.1 Geometry and operating states

The full machine has six poles and 36 stator slots. We model a 60° sector, polar angles 60° to 120° at zero rotation, with antiperiodic conditions on the radial boundaries. The rotor (radii 16 to 44 mm) carries one buried permanent magnet per pole and turns; the stator (radii 45 to 67.5 mm) is fixed and carries the windings in the slots between its teeth. Axial length is $L = 0.1$ m. The 1 mm air gap is split at the coupling radius $r_c = 44.5$ mm, where the separately meshed rotor and stator meet. Figure 1 shows the initial design with its parts, the eight scalar design parameters that form the smaller design space, S8, and the 55 control points whose grouped offsets the larger space, S18, adds (both defined in Section 5).

The reference describes the geometry as NURBS patches, spline pieces defined by weighted control points, and the reconstruction uses its initial exports. Their control points are stored with weights already multiplied in, and patch orientation must be accounted for when integrating areas. Two thin rotor patches described as air slits in source comments receive iron under the reference's material assignment; we follow that assignment. The mesh generator, built on Gmsh 4.15.2 (Geuzaine and Remacle, 2009), keeps the 12 winding groups (the two coil sides of each slot)

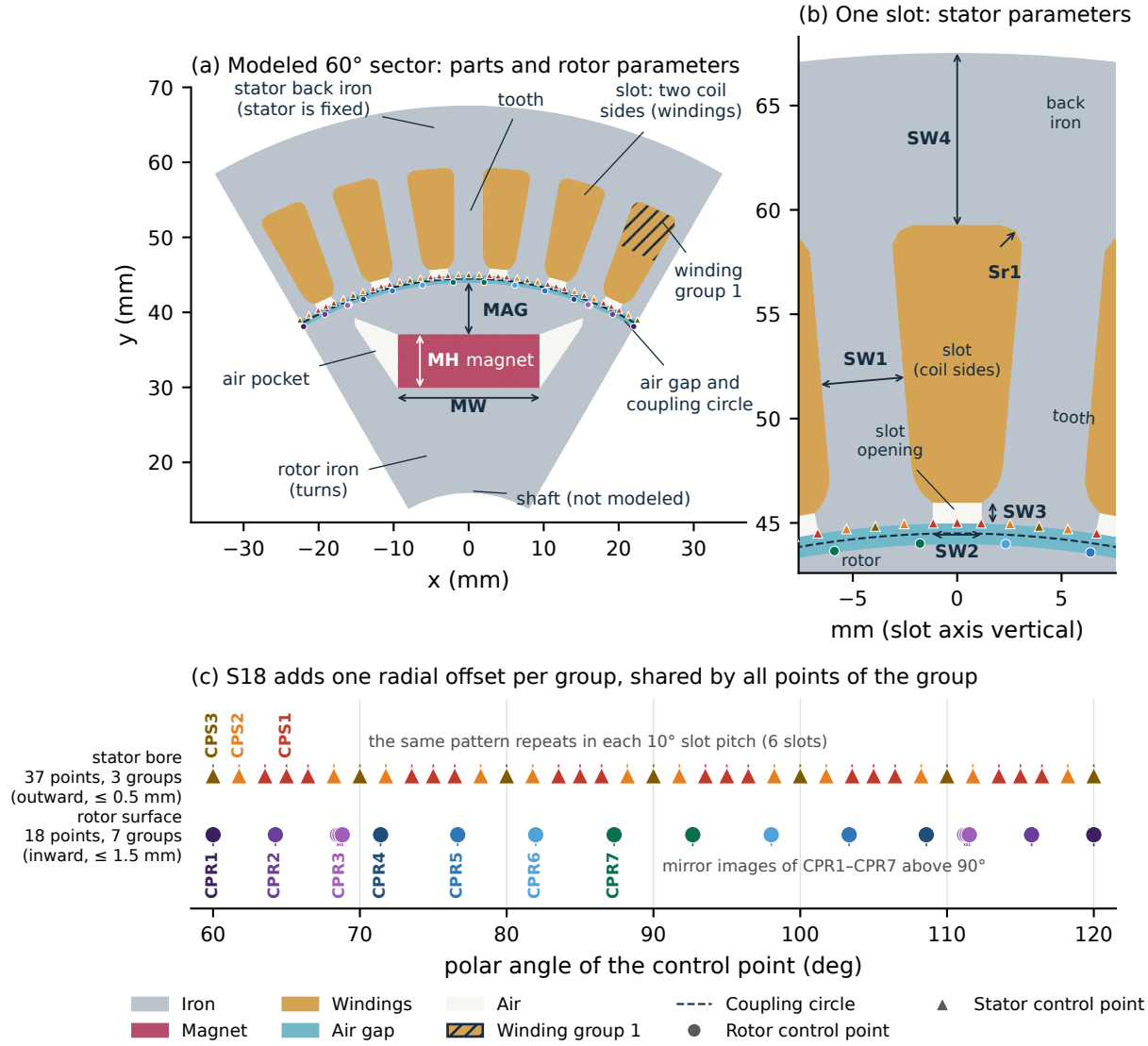


Figure 1: Initial design, rendered from the committed coarsest mesh (element edges hidden). (a) The modeled sector and its parts, with the rotor parameters: MAG, iron depth between the magnet and the rotor surface on the pole axis; MH, magnet height; MW, magnet width. Winding group 1, whose current density the reference optimizer holds fixed (Section 4), is hatched. (b) One slot, turned so that its axis is vertical, with the stator parameters: SW1, tooth width; SW2 and SW3, width and depth of the slot opening; SW4, back-iron depth on the slot axis; Sr1, radius of the fillets at the slot bottom. (c) The 55 control points on the rotor surface and the stator bore, by polar angle. The 18-parameter space S18 (Section 5) adds ten offsets, one per group (CPR1–CPR7 on the rotor, CPS1–CPS3 on the stator); an offset moves every point of its group radially by the same distance, inward on the rotor (at most 1.5 mm, 1 mm for CPR3) and outward on the stator (at most 0.5 mm). Markers in (a) and (b) use the colors of (c).

separate, since their two characteristic areas are approximately 36.29 and 31.22 mm^2.

At mechanical angle β, phase currents are

$$I_q(\beta) = aI_* \sin(\phi_0 + 3\beta + \delta_q), \qquad (\delta_U, \delta_V, \delta_W) = (0, -2\pi/3, 2\pi/3), \tag{1}$$

where the current fraction $a \in \{0, 0.5, 1\}$ selects three operating states (at $a = 0$ only the magnets act; $a = 0.5$ and 1 are loaded), $I_* = 9.677657374\,\mathrm{A}$, and the electrical phase offset $\phi_0 = 0$ for the initial design. Angles in Eq. (1) are in radians. Each winding group carries current density $J_g = 12 s_g I_q / |\Omega_g|$, with its own area and prescribed sign $s_g \in \{-1, 1\}$. Using a common area would change the current density between groups by about 16%. We compare 20 distinct mechanical angles, $0°, 1°, \ldots, 19°$, excluding the duplicated endpoint in the source CSV. The reference's optimization grid contains the ten even angles; both grids are retained in the artifacts.

3.2 Field model and rotor–stator coupling

We solve two-dimensional nonlinear magnetostatics for the vector potential A_z with continuous piecewise-linear triangles and a tabulated M27 iron law, using Newton's method with the consistent material tangent. The separately meshed rotor and stator meet on the coupling circle, across which the rotor mesh turns, in one of two ways. A conforming interface identifies matching trace nodes after rotation. The harmonic model, following the reference, matches the two sides only in the Fourier modes of the field on that circle of orders $n = 3, 9, \ldots, N_h$, through Lagrange multipliers, with $N_h = 45$ unless stated otherwise. It passes only the lower orders of the interface field, like a low-pass filter; a higher interface order N_h passes more of them, and the conforming interface passes all that the mesh resolves. Torque is evaluated either by the Arkkio expression, the Maxwell stress averaged over the air-gap annulus (Arkkio, 1987), or, for the harmonic model, by the coupling torque, the derivative of the stored energy with respect to rotation through the interface constraint (Egger et al., 2022). The forward comparisons of Section 3.4 use Arkkio for both couplings to isolate the interface change; the derivative study of Section 4 pairs harmonic coupling with coupling torque and conforming coupling with Arkkio; and the optimization study of Section 5 runs its optimizers on order-45 coupling torque (its *coarse simulation*) and scores them at order 93 (its *validation simulation*). Retaining every order below the trace Nyquist limit reproduces the conforming Arkkio torques within $4.1 \times 10^{-9}\,\mathrm{N\,m}$ and $1.3 \times 10^{-9}\,\mathrm{N\,m}$ on the two coarsest meshes, L0 and L1 (Section 3.3), without fitting any parameter to the saved torque. Appendix A gives the weak form, tangent, iron law, coupling constraints and torque expressions.

3.3 Meshes and statistics

The study uses four meshes, L0 (coarsest) to L3 (finest), with 3,153, 9,937, 36,083 and 139,268 nodes. The 60° coupling arc has $120 \cdot 2^{\ell}$ segments at level ℓ, so integer-degree rotation is represented at every level. A solution at one angle initializes the next angle at the same current. We report the mean $\bar{T}$ and the population standard deviation σ_T of the torque over the $N = 20$ angles; σ_T is the ripple metric. At zero current it measures cogging, the torque variation caused by the permanent magnets and slots without applied phase current. The squared-ripple objective is $F = \sum_{a \in \{0, 0.5, 1\}} \sigma_T(a)^2$, in $\mathrm{N^2\,m^2}$, on the 20-angle grid. Torque ripple is a source of vibration and acoustic noise, and the reference optimization minimizes it; because a design can also lower ripple by giving up torque, the optimization study holds a floor on the mean torque (Section 5). Appendix A lists the implementation checks.

3.4 Interface order sets the validation simulation

Table 1 compares the same geometry, material and excitation with two interface models. The conforming L3 means are 0.15% and 0.16% below the saved reference at half and full current.

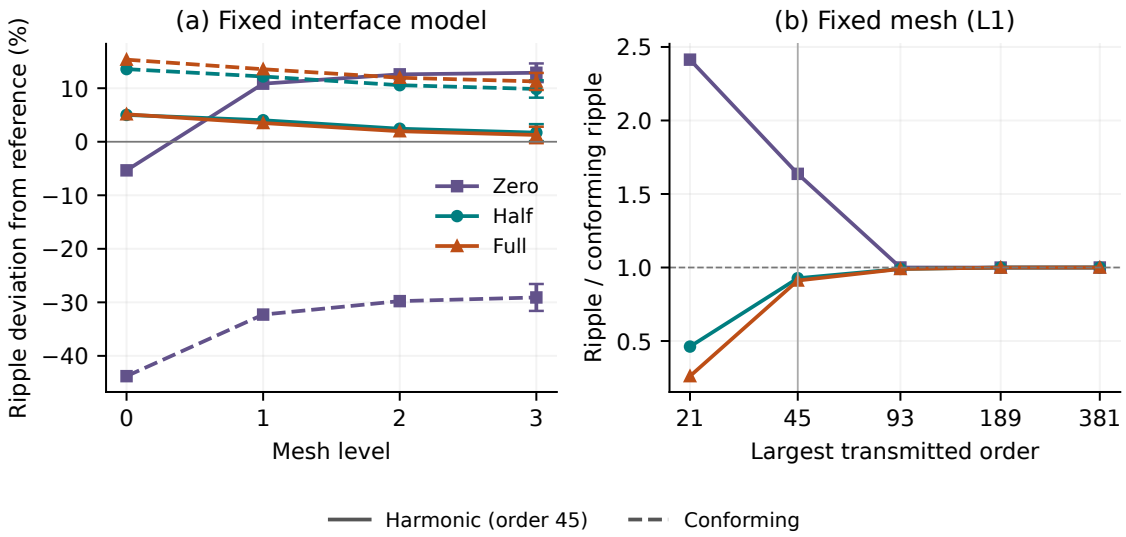


Figure 2: Left: ripple deviation from the reference under spatial refinement, with solid lines for harmonic order 45 and dashed lines for conforming coupling; L3 bars use Eq. (11). Right: order refinement on L1 with logarithmic order spacing, normalized by the conforming ripple at that same level. The saved reference uses order 45; see Section E.1 for its refinement. Colors identify current, not coupling.

Harmonic coupling reduces the mean deviations to -0.002% and $+0.051\%$, with both reference means inside the estimated refinement intervals. The loaded ripple deviations fall from $+9.9\%$ and $+11.3\%$ to $+1.7\%$ and $+1.3\%$.

Table 1: L3 population standard deviation and mean torque, in N m per sector, on the 20-angle grid. The last column gives ripple deviation from the saved reference; n/a identifies the reference itself. The reference is subsequently refined in Section E.1.

Current fraction	Model	Mean	Std	Std vs ref. (%)
0	Reference	0.00010	0.01071	n/a
0	Conforming	0.00001	0.00760	−29.1
0	Harmonic	0.00000	0.01209	+12.9
0.5	Reference	0.66208	0.03895	n/a
0.5	Conforming	0.66106	0.04279	+9.9
0.5	Harmonic	0.66206	0.03961	+1.7
1	Reference	1.44248	0.07935	n/a
1	Conforming	1.44023	0.08829	+11.3
1	Harmonic	1.44322	0.08036	+1.3

Spatial refinement at fixed order and order refinement on a fixed mesh answer different questions (Fig. 2; Appendix E defines the refinement interval). At L3, order 45 raises cogging standard deviation by about 59% and lowers loaded standard deviation by 7–9%, relative to the conforming values. At L1 and L2, order 189 approaches the conforming torque to within 1.1×10^{-4} N m. A spatially refined order-45 solution thus remains a solution of a truncated interface model. Matching that truncation improves reproduction of the published calculation; it does not establish which model better represents the physical motor.

Rerunning the public reference with its own isogeometric model at four refinement levels reduces its cogging discrepancy with GEODE from 12.9% to 0.10%, so the coarse saved reference explains

the initial cogging gap at matched order, although neither refinement sequence is fully converged. For the reference's saved optimized design (its iteration-120 design, Opt120), raising the order from 45 to 93 multiplies its squared-ripple objective by 3.6. Appendix E gives these checks, the estimator and material controls, and the saved design's size and current normalizations. For the optimization study (Section 5) this sets the validation simulation: its optimizers search at order 45, and a design is credited only after it is rescored at order 93 on a finer mesh, so that a reduction that exists only in the truncated model does not count.

4 Design-gradient verification

4.1 Differentiated model and excitation policy

For a design vector p, the converged state z contains the reduced potential and, for harmonic coupling, its multipliers. Writing the full constrained residual as $R(z, p) = 0$, the discrete adjoint is

$$R_z^T \psi = F_z^T, \qquad \frac{\mathrm{d}F}{\mathrm{d}p} = F_p - \psi^T R_p. \tag{2}$$

We differentiate mesh coordinates, element terms, magnet sources, current loads, harmonic constraints and the torque functional. The explicit coordinate derivative of Arkkio is included for conforming coupling. Fixed-topology mesh motion $X(p)$ supplies the chain rule $\mathrm{d}F/\,\mathrm{d}p_k = \langle \mathrm{d}F/\,\mathrm{d}X, \partial X/\partial p_k \rangle$ for geometric parameters, with direct terms added where needed.

The 12 parameters comprise nine geometric lengths (MAG, MH, MW, SW1–SW4, SD1 and Sr1), axial length, operating angle and radial scale. The two designs are the initial parameters and an interior point halfway toward the saved optimized scalar parameters (Appendix J). This verifies a local scalar parametrization. The reference's optimized geometry also uses grouped surface-control displacements; reproducing its 12 scalar values alone does not reproduce that geometry. The gradients that the optimizers of Section 5 use, including the grouped offsets of S18 and its mesh-motion map, are checked separately there.

The excitation rule changes these derivatives. Holding peak current fixed differs from the reference optimizer's rule, which holds winding-group-1 current density fixed (Appendix H defines both). At L1 (harmonic coupling, coupling torque, full current), the mean-torque derivative with respect to the stator tooth width SW1 is $+1.68\,\mathrm{N\,m/m}$ under fixed current and $-220.85\,\mathrm{N\,m/m}$ under fixed group-1 density (Fig. 3): a wider tooth narrows the slot and lowers current under the density rule. At this state the policy changes seven of the ten geometric/scale derivatives and reverses the mean-torque signs for SW1, SW4 and SD1. The fixed-size problem of Section 5 holds physical current fixed for every method.

4.2 Finite differences and Taylor remainders

The derivative objectives are mean torque and population standard deviation on the ten even mechanical angles, matching the reference's optimization grid. Central differences move the same mesh at $p \pm he_k$; step sweeps separate truncation from floating-point and nonlinear-solve error. We also check the first-order remainder $|F(p + he_k) - F(p) - h\ \mathrm{d}F/\,\mathrm{d}p_k|$ for quadratic decay.

Across initial and perturbed designs, L0–L2 and the recorded loaded current/rule subsets, there are 800 objective–parameter records. They pool harmonic/coupling-torque and conforming/Arkkio evaluations. The fixed-density MAG, MH and MW records repeat 96 fixed-current comparisons byte-for-byte, leaving 704 distinct comparisons. Of these, 688 have a best-step relative error at most 2.8×10^{-7}; the maximum is 1.07×10^{-5} for a small Sr1 ripple derivative. This is a best-step verification statistic, not the accuracy of a finite-difference step chosen in advance. Figure 3 includes the 704 distinct cases and an explicit Taylor sweep. The artifact retains every step, rejected

displacement and relative error. The recorded subsets do not form a full factorial study. Of 592 eligible distinct loaded comparisons, 530 have a quadratic Taylor-remainder window spanning at least four decades of step size. Verified is not the same as model-independent: at the same design the two model–estimator pairs give ripple sensitivities whose box-scaled vectors differ by 6.7% and 10.2% of their norm, more than either changes under mesh refinement. Appendix H gives the Taylor windows, the weaker zero-current checks, mesh convergence and this model dependence.

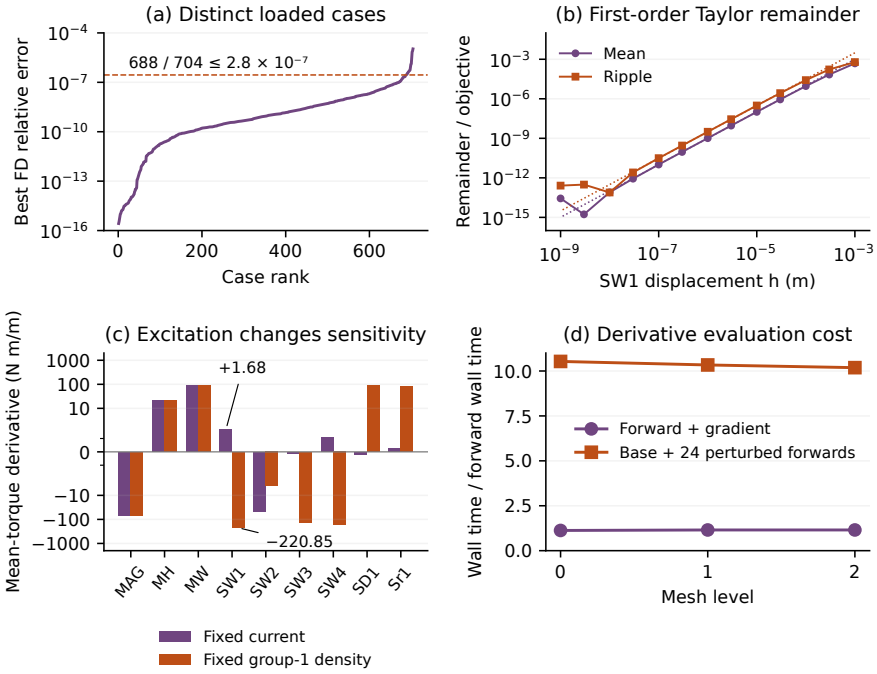


Figure 3: Discrete gradient evidence. (a) The 704 distinct loaded best-step finite-difference errors, sorted, pooling harmonic/coupling-torque and conforming/Arkkio objective–parameter cases. (b) SW1 Taylor remainders at L1, initial design, full current, harmonic coupling and coupling torque, fixed current; the plotted remainders are divided by $|F(p)|$, and dotted h^2 guides are anchored at $h = 10^{-5}$ m. (c) Mean-torque sensitivities for the nine geometric parameters under two current rules at that same state; the vertical axis is symmetric-log, linear within ± 2 N m/m. (d) Recorded cost ratios for harmonic coupling at tolerance 10^{-14}; gradients include the forward evaluation, and FD includes its shared base forward. L0/L1 use medians of three runs, L2 a single run; the seconds are in Appendix H. These timing data were collected under uncontrolled host load.

On one loaded host, across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient of both aggregate objectives with respect to all 12 parameters costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve (Fig. 3d; Appendix H gives the protocol and seconds). A forward here is one ten-angle sweep at full current on a multithreaded host, not the 60-solve, single-core forward charged in the optimization study (Section 5); the two studies' cost ratios are not interchangeable.

5 Matched optimization study

This section tests whether adjoint-driven optimization shortens the time needed to reach a common validated ripple target. The protocol and all of its amendments were fixed in the preregistration before any main-run outcome was observed. One later deviation, in how the final refinement was run,

was adopted after the main runs had finished and is described in the *Final refinement* paragraph of Section 5.2; it changes no selection rule, check or target.

5.1 Protocol

Coarse and validation simulations. The optimizers never query the simulation that scores them. Each queries a *coarse simulation*, the L1 mesh (the coarser L0 for methods D and E) with interface order 45, and its search is driven only by coarse values. Every design that becomes the method's new best, its incumbent, is rescored on one *validation simulation*, the finer L2 mesh with interface order 93, which no optimizer sees; only validated scores count toward a target. Both are physics simulations, the nonlinear finite-element model of Section 3 at coarser and finer mesh levels and interface orders, neither fitted to torque data nor corrected against validation; of what the optimizers use, only their own search models, C's radial-basis surrogate and COBYQA's interpolation model, are fitted, to coarse values. Validation is charged to the run that requests it, as in a design loop that must check its candidates before trusting them; this validate-every-incumbent rule costs methods that change incumbents often the most, and Section 5.2 reports its share of each run's time.

Methods. Five methods form an optimizer-by-resolution comparison plus one surrogate (Table 2). A and E are SLSQP (Kraft, 1988) using the field adjoints of Section 4, safeguarded so that a refused request restores the last admissible point and restarts the solver; B and D are COBYQA (Ragonneau, 2022) with $2n + 1$ interpolation points and unit-box trust radii 0.1 to 10^{-6}, where n is the number of free parameters. Both run through SciPy 1.18.1 (Virtanen et al., 2020). C is an empirical trust-region method on four cubic radial-basis interpolants of the L1 coarse outputs (the three per-current standard deviations and the full-current mean), refitted after every evaluation from an $n + 1$-point initial design; Appendix B gives its radii, acceptance and update rules and the other settings. B against A and D against E vary the optimizer at fixed coarse resolution, and E against A and D against B vary resolution at fixed optimizer; every paired ratio below is written in this order (B/A, D/E, E/A, D/B), so a value above one favors the second method. The optimizer contrast changes the search strategy and the gradient source together; Section 6 separates them with the three further methods of Table 2. The coarse simulation uses order-45 coupling torque on 20 angles at current fractions 0, $\frac{1}{2}$ and 1 of the fixed peak current 9.6777 A. L0 and L1 differ only in spatial resolution. Time to a fixed target, with unattained targets kept as censored, follows the data- and performance-profile practice for benchmarking optimization solvers (Dolan and Moré, 2002; Moré and Wild, 2009). The preregistration also named a fixed group-1 current-density rule as a labelled control. It was not run in this study, which uses the fixed physical current throughout; Section 4 reports that rule's effect on sensitivities.

Table 2: Methods. A–E form the preregistered study (two repeats each). FD-cold, FD-warm and A′ form the post-hoc ablations of Section 6; each ablation ran in its own session with A rerun unchanged as A′.

Method	Optimizer or search model	Gradient	Coarse simulation
A	Safeguarded SLSQP	Field adjoint	L1
B	COBYQA	None	L1
C	Empirical cubic-RBF trust region	None	L1
D	COBYQA	None	L0
E	Safeguarded SLSQP	Field adjoint	L0
FD-cold	Safeguarded SLSQP	Forward differences, cold stencils	L1
FD-warm	Safeguarded SLSQP	Forward differences, warm stencils	L1
A′	A, rerun in each ablation	Field adjoint	L1

Design spaces and constraints. The 8-parameter space S8 frees eight geometric lengths: MAG, MH, MW, SW1–SW4 and Sr1 (Fig. 1a,b), that is, the rotor iron depth, magnet height and width, four stator tooth and slot dimensions, and the slot-bottom fillet radius. The 18-parameter space S18 adds the ten grouped offsets of Fig. 1c, which move 55 control points of the rotor surface and the stator bore. Stack length, radial scale, stator outer diameter and operating angle are fixed, so the saved Opt120 design lies outside both spaces. Table 3 compares these spaces with the 12-parameter set whose gradients Section 4 verifies. The two spaces also use different discrete mesh-motion maps, so their contrast does not isolate the ten added offsets. Each coarse simulation $r \in \{\mathrm{L0}, \mathrm{L1}\}$ minimizes $F_r = \sum_c \sigma_{r,c}^2$ over the three objective currents c, subject to $\bar{T}_{r,1} \geq 1.0025\,\bar{T}_{r,1}(\theta_0)$ for its own full-current mean and to normalized construction margins $c_i - 0.05 \geq 0$. Here θ_0 is the reference's published initial design. Unbuildable or mesh-infeasible candidates are refused: SLSQP restores toward the last admissible point and restarts, COBYQA receives a NaN barrier, and C skips the point. A moved mesh with an inverted triangle or a minimum angle below 5° is repaired by up to four charged remeshing attempts. Because refinement cannot remove a limiting geometric corner, the 5° policy also acts as an implicit geometric constraint. S18's slot-opening channel wall is not guarded and can become very thin at extreme offsets; that risk is handled by repair and refusal, not by a wall-thickness constraint.

Table 3: The three parameter sets. Section 4 verifies the gradients of a 12-parameter scalar set; the optimizers search S8 and S18, whose gradients are checked separately (Appendix C). Values are in Table 17.

Parameters	Gradient check (12)	S8	S18
MAG, MH, MW (magnet)	free	free	free
SW1–SW4, Sr1 (stator slot)	free	free	free
SD1, axial length, ϕ_0, radial scale	free	fixed	fixed
CPR1–CPR7, CPS1–CPS3 (offsets)	—	—	free

The gradients A and E use, including those of S18's grouped offsets, agree with central differences in both spaces within 2.8×10^{-7} relatively in the directional checks (Appendix C).

Common validation and credit. Every method is judged on the same order-93, L2, 20-angle validation simulation. Its initial-design anchor is $F_0 = 9.6034 \times 10^{-3}\,\mathrm{N^2\,m^2}$ with full-current mean $\mu_0 = 1.44084\,\mathrm{N\,m}$. Target T_x is an x% reduction of F_0: $T_{50} = 0.5F_0$, $T_{90} = 0.1F_0$ and $T_{99} = 0.01F_0$. The *primary targets* are T_{90} in S8 and T_{99} in S18, the deepest target reached in each space (no run

reaches T_{99} in S8). A validated design is credited, and attains a target, only if

$$F_{\mathrm{val}} + \delta F \le T, \qquad \delta F = \max\{0.10 F_{\mathrm{val}},\, 2.5 \times 10^{-7}\,\mathrm{N}^2\,\mathrm{m}^2\}, \tag{3}$$

its validated full-current mean is at least $1.0025\,\mu_0$, its construction margins and mesh checks pass, and its validation completes within budget. The allowance δF is an empirical margin, not a bound on continuum error. Optimizer success flags confer no credit. Changed incumbents are validated as they appear, after the initial interpolation set for B, C and D. A run's *target crossing* is its first validated design that attains the target; the time to the target is the charged time at which that validation completes.

Budget, timing and cost vocabulary. Cost is wall time, in units of the median cold validation forward measured on the same host at the same concurrency (66.66–70.94 s in the three sessions). A *forward* is one evaluation of the objective: 60 nonlinear field solves, one per current fraction (3) and rotor angle (20). In a *cold* forward, Newton starts from zero at the first angle of each current and from the previous angle's solution after that; in a *warm* forward, used only by FD-warm's stencils (Section 6), every field solve starts from the base point's converged field at the same current and angle. A finite-difference gradient adds one *stencil solve*, a coarse forward at a perturbed point, per parameter. Table 4 gives what each operation costs in this unit. Each run receives 120 units, with a 60-unit checkpoint that counts only validations completed by then. The charge covers evaluator start-up, geometry checks, mesh motion and repair, coarse forwards, adjoints, optimizer overhead, surrogate fitting and validation; nothing is subtracted as overhead. L0 receives no discount.

Table 4: Cost vocabulary: medians of the recorded charges, in units of the session's cold validation forward. The validation charge is that of the 3,350 completed validations of the main study; the other rows are from the ablation sessions (Section 6), for $n = 8$ and $n = 18$ parameters.

Operation	Simulation	Cost (units)
Validation forward	L2, order 93	0.87 (range 0.55–1.10)
Coarse forward	L1, order 45	0.216–0.245
Adjoint gradient, all n components	L1, order 45	0.054–0.061
FD-cold gradient: n cold stencil solves	L1, order 45	1.90 and 4.25
FD-warm gradient: n warm stencil solves	L1, order 45	0.97 and 2.36

Starts and execution. Each space uses θ_0 and four Latin-hypercube starts (LHS 1–4, seeds 1–4), with the scalar coordinates matched across spaces. A start was eligible only if it was buildable and admissible, and its moved reference mesh met the mesh policy without repair, in both spaces at L0 and L1. The starts therefore favor moderate deformations and are not uniform samples of the design box. All starts were frozen before any main outcome. Every method runs from every start twice, giving 100 runs. Within a start, repeat 1 rotates the method order and repeat 2 reverses it.

Each run is one single-threaded process; 15 ran concurrently on one AWS c7a.4xlarge host, and all 100 completed in one session without recovery or host contamination (Appendix C).

5.2 Results

Repeats. The two repeats of each run follow the same optimization path. In all 50 pairs, the sequences of validated objectives agree exactly up to the shorter run's length, and they differ in length only when the budget or a solver stop cuts one repeat earlier. The repeats therefore measure timing noise, not independent outcomes, and the comparison is a paired case study over five starts.

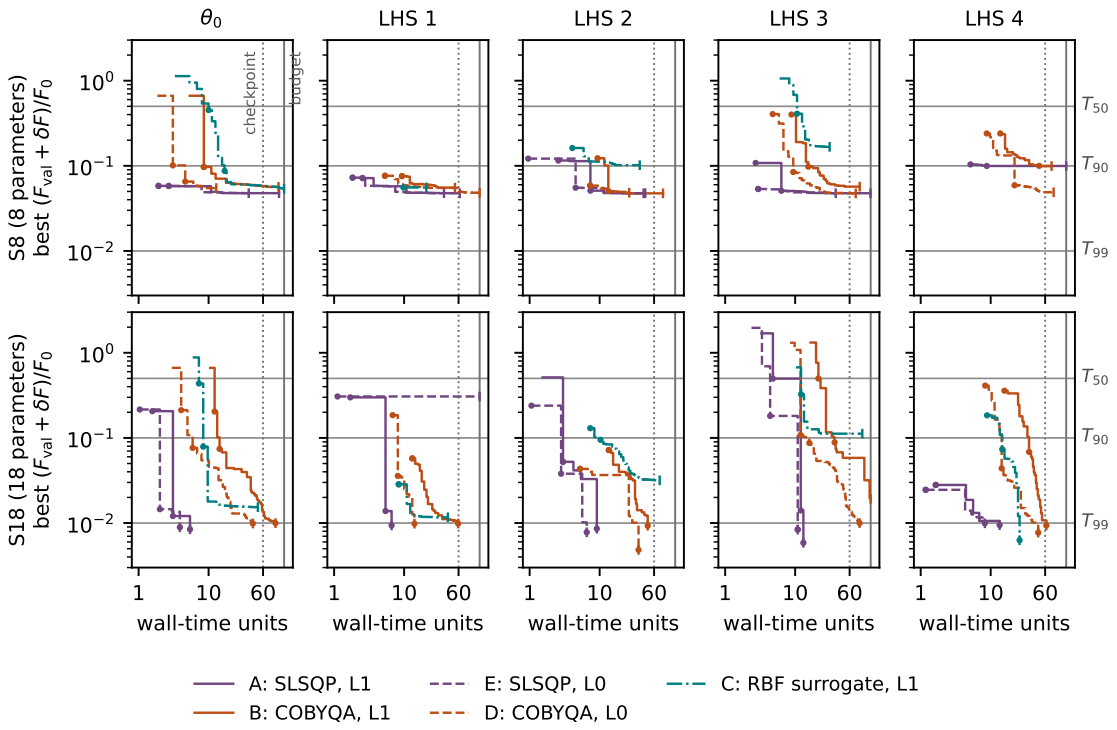


Figure 4: How fast each method improves. Each panel is one start (θ_0 or Latin-hypercube start LHS 1–4) in one space; each curve is one method's best credited $(F_{\text{val}} + \delta F)/F_0$ so far, that is, its validated objective plus the allowance δF, against charged wall time (repeat 1, log axes). Lower is better. The labelled grey lines are the targets T_{50}, T_{90} and T_{99}, and a dot marks a curve's first crossing of each; the dotted and solid verticals are the 60-unit checkpoint and the 120-unit budget. Each curve ends at the run's actual stop (tick); a missing curve means the run had no credited validation.

Over the 111 target crossings attained in both repeats, the time differs by a median of 2.1% and at most 12.4% relative to repeat 1. Repeat-1 times are reported below.

Attained quality. Figure 4 shows the best credited value of $(F + \delta F)/F_0$ for every start, method and space (repeat 1). In S8, no run goes below $F/F_0 = 0.043$ and none reaches T_{99}. The best designs of A, B, D and E all reach $0.043F_0$, and C's reaches $0.049F_0$; in Fig. 4, which includes the allowance, the A, B, D and E level appears near 0.048 (C's near 0.054). All five lie within 0.17% above the full-current mean floor, so the mean constraint is active there: the level shared by A, B, D and E is the constrained optimum they found in the eight-parameter space, not a property of any one method. In S8 the start mattered more than the method. From LHS 4, A and B stop near $0.09F_0$, D reaches $0.044F_0$, and C and E never produce a credited validation, in either repeat. In S18, each method's best design reaches 0.004–$0.009F_0$, and every method except C reaches T_{99} from at least four of five starts.

Time to target. Figure 5 gives the charged time to each target, per start, and Table 7 in Appendix D lists the values. In repeat 1, only three crossings occur after the 60-unit checkpoint, all at T_{99} in S18: two by B and one by D.

Table 8 (Appendix D) gives paired time ratios. A ratio is formed only when both methods attain the target from that start; no finite ratio is derived from a censored cell. Wherever both reach a

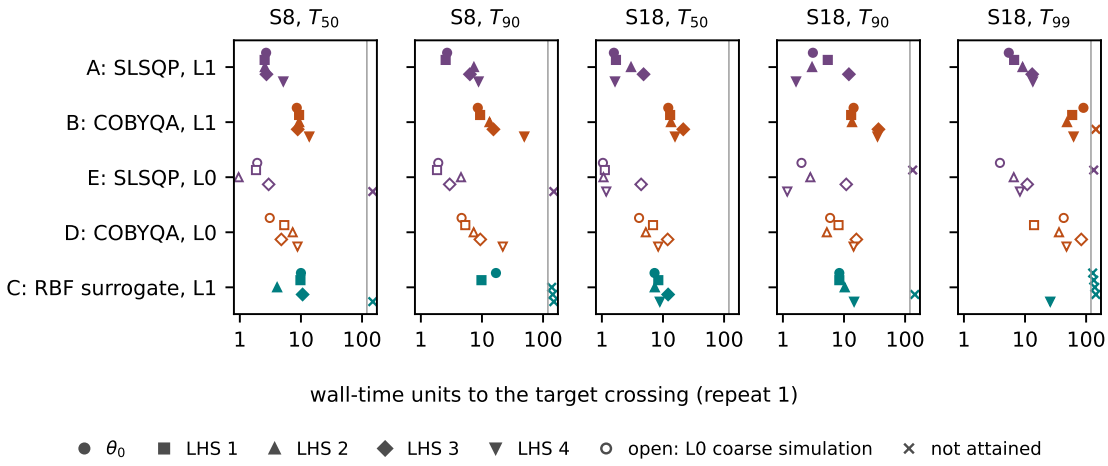


Figure 5: Charged wall time to each target crossing, repeat 1, for every target attained by some run. Each row is a method, and each marker shape is one start, offset vertically so that the same start can be followed across methods. Filled markers: L1 coarse simulation; open markers: L0; crosses: not attained within 120 units.

target from the same start, gradient-based SLSQP gets there sooner than COBYQA at the same coarse resolution. At L1 (B/A), the geometric-mean ratio is 3.10–3.25 in S8 and 5.06–7.79 in S18. At L0 (D/E), it is 2.45–2.75 and 3.14–7.20. All 45 finite optimizer-contrast pairs (by space, target and start) favor SLSQP, with ratios from 1.46 to 21.5. The coarse mesh shortens the time by a smaller factor: the geometric-mean E/A ratio is 0.62–0.79 and D/B is 0.45–0.56.

Validation charges. Validating every new incumbent makes validation the largest single charge (Table 10 in Appendix D). One validation costs a median of 0.87 units over the 3,350 completed validations (range 0.55–1.10), and validation takes 53% of all charged time, 2–90% per run. SLSQP crosses a target after few validations, but most of its time to a target is validation; COBYQA at L1 (B) needs more validations, which are a smaller part of its time. Excluding the validation charges up to each crossing, the same-resolution geometric means rise to 5.41–5.66 (B/A) and 5.55–5.56 (D/E) in S8, and to 8.84–13.58 and 7.92–15.27 in S18, so the ratios as charged are conservative for the optimizer contrast. The coarse mesh cuts the remaining time more, by geometric means of 0.19–0.27 (E/A) and 0.20–0.23 (D/B), but it does not make validation cheaper, which is why its overall saving is only 1.3–2.2 times. Against D, the derivative-free competitor on the coarser mesh, A's lead in S8 comes mostly from validation: D needs more validated incumbents before it crosses (a median of six against two), and each costs a full validation. As charged, A is faster than D by a geometric mean of 1.68–1.83 in S8; excluding the validation charges it is 1.15–1.16, and two S8 pairs then favor D (Appendix D). In S18 A's lead is 2.25–4.32 as charged and 1.92–3.19 without validation. This contrast changes both the optimizer and the resolution; one of its 25 pairs is a near tie (0.99, S8 LHS 2 at T_{90}: D is 0.75% faster, within the timing noise between repeats).

The speed of E has a reliability cost. E never produces a credited validation from S8 LHS 4. From S18 LHS 1 it reaches T_{50} but then stalls at $0.28F_0$, missing T_{90} and T_{99}. In four start–target cells, D reaches a target that E misses. A, by contrast, reaches every target that any method reaches, from every start. C attains T_{99} only once in S18 and reaches T_{90} from two of five S8 starts.

Final refinement. The preregistered selection takes, for each method and space, the lowest-F_{val} feasible design over both repeats, and the earliest repeat-1 crossing of each attained target. With θ_0 in each space and identical designs merged, this gives 33 designs. Each received the six registered checks: the primary L2 validation, L3, a 40-angle grid, Newton tolerance 10^{-12}, an independently solved grid shifted by 0.5°, and held-out currents 0.25 and 0.75. The 15 designs with zero stator offsets also received the conforming outer-band Arkkio check on fresh meshes. All 213 checks completed (Table 9 in Appendix D). These checks qualify the selected designs only; they do not check each of the study's 111 repeat-1 target crossings.

This is the one deviation from the preregistration noted at the start of Section 5: the registered pipeline began by capturing the closed study on its original host, which had been restarted, so the refinement used a hash-verified copy of the study outputs on a different host (Appendix C). Spatial refinement is the largest change. Over all 33 designs, L3 moves F by -4.0% to $+6.3\%$. Over the 25 target crossings alone, the range is -4.0% to $+1.1\%$. The largest increase belongs to the overall best design, D's in S18, whose F rises from $0.0044F_0$ to $0.0047F_0$. Angular and nonlinear effects are small: the 40-angle and shifted grids change F by at most 0.11%, and the tighter Newton tolerance by less than 1.1×10^{-5} relative. The conforming estimator gives 1.1% to 4.4% more ripple than the order-93 coupling on the S8 designs, and 2.0–2.1% more at θ_0. All 25 target crossings still satisfy their target, allowance included, at L3; the smallest remaining margin is 1.2% of the target. Every candidate also satisfies the full-current mean floor in the primary, L3, 40-angle, tight and shifted checks, the closest by 0.012%. The refinement therefore confirms the selected crossings and leaves the time-to-target comparison unchanged. At held-out currents 0.25 and 0.75, part of S18's extra reduction is lost (Appendix D).

6 Post-hoc ablations: adjoint against finite-difference gradients

The contrast of Section 5 changes the search strategy and the gradient source together. Two ablations change only how SLSQP obtains its gradient, and therefore what the gradient costs. Both were added after the main outcomes were known and are not part of the preregistered study; each public protocol was frozen before any of its runs (Code and data availability). We call the methods of an ablation its *arms*.

FD-cold. FD-cold is the safeguarded SLSQP of A with identical bounds, constraints, restoration, validation policy, targets, mean floor and stopping rules. Only the gradients of the objective and of the mean constraint change: they are forward differences with a step of 10^{-6} in each coordinate's physical unit (metres for the lengths and offsets), reusing the base value, so one gradient costs n extra coarse forwards, the stencil solves ($n = 8$ in S8 and 18 in S18). Each stencil solve is an ordinary coarse forward on the base point's mesh generation, and like every coarse forward in the preregistered coarse configuration it starts Newton-cold (all 40 traces record a cold start). A component whose forward step is unusable (outside the box, unbuildable or on another mesh generation) takes the backward step, as 8% of components did; no gradient was refused.

FD-warm. FD-warm, the second arm, is FD-cold except that each stencil solve starts Newton from its base point's retained converged state on the same mesh generation, which makes it the stronger finite-difference competitor. In both arms every solve is charged, and the geometry-constraint Jacobians remain analytic. Every run is one single-threaded process, so the n stencil solves run one after another and the time ratio is also a ratio of compute; solving the stencil in parallel would shorten the wall time per gradient but not its compute.

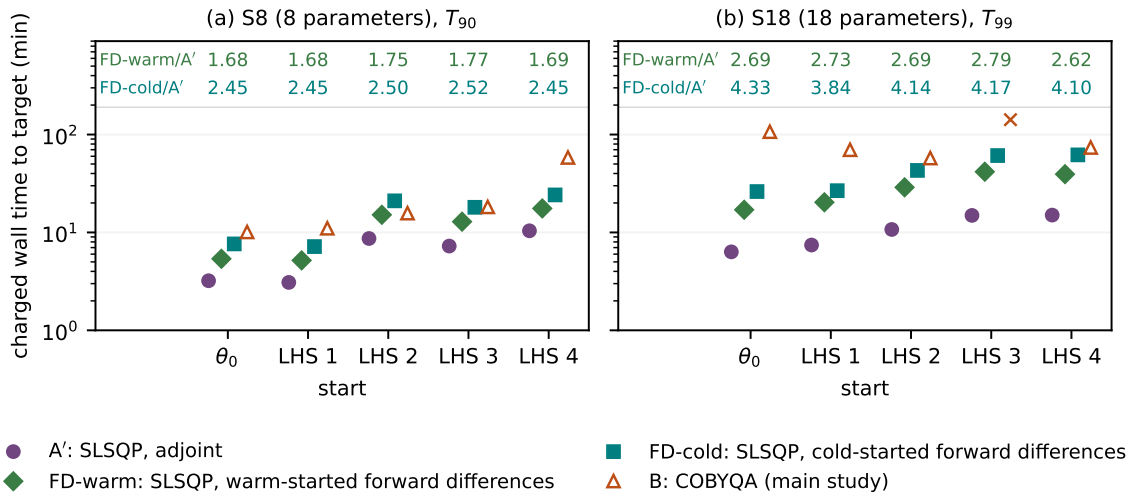


Figure 6: Charged wall time, in minutes, to the primary target from each start (repeat 1): A′ (SLSQP with adjoint gradients, from the FD-warm session), FD-warm and FD-cold (the same SLSQP with warm- and cold-started forward-difference gradients, from their own sessions), and B (COBYQA, main study). Minutes avoid the 3–6% differences between the sessions' time units (1 unit is 1.11 to 1.18 minutes). The cross marks the start from which B did not reach T_{99}; it is drawn at B's 120-unit budget. The two labelled rows above the line give each start's FD-warm/A′ and FD-cold/A′ ratio of charged time, each formed within its own session: FD-warm/A′ uses the plotted A′, and FD-cold/A′ uses the FD-cold session's A′, which is not plotted and agrees with the plotted one within 4% (Appendix D).

Controls and estimand. Each arm runs with A rerun unchanged in the same session (A′), sharing host, time unit and load, from the five starts in both spaces on the L1 coarse simulation at 120 units: FD-cold with two repeats (40 runs), FD-warm with repeat 1 only (20 runs), as its protocol fixed in advance because FD-cold's repeats had agreed to about 1% (a median 1.0% per primary crossing time) and the estimand uses repeat 1. The primary estimand is, per start, the repeat-1 time ratio to the primary target, T_{90} in S8 and T_{99} in S18; a censored cell gives no finite ratio.

Results. Both finite-difference arms and both A′ controls reach T_{50} and T_{90} in S8 and all three targets in S18 from all five starts; none reaches T_{99} in S8, and no cell is censored. Against the stronger baseline, the adjoint reaches the primary target sooner from every start (Fig. 6): the geometric-mean FD-warm/A′ ratio is 1.71 in S8 (range 1.68–1.77) and 2.70 in S18 (2.62–2.79). Against cold-started stencils it is 2.47 (2.45–2.52) and 4.11 (3.84–4.33), with repeat 2 giving 2.49 and 4.04 (Table 11); compared across sessions, FD-warm takes about 0.72 and 0.68 of FD-cold's time in seconds. All three arms cross each primary target after the same number of validations from every start, with one exception at a secondary target (Appendix D), so each gap is mainly the cost of the gradient. Excluding the validation charges up to each crossing, FD-warm/A′ becomes 3.23 and 6.10, and FD-cold/A′ 5.58 and 10.07.

Gradient factor and optimizer factor. Against the main study, B/A $\approx$ (B/FD-warm)(FD-warm/A′) splits the SLSQP-versus-COBYQA ratio into a gradient factor, finite-difference against adjoint SLSQP, and an optimizer factor, COBYQA against finite-difference SLSQP. Table 5 gives both splits over the starts where B attains the primary target (all five in S8; four in S18, where B misses LHS 3). Over those four starts the FD-warm/A′ and FD-cold/A′ means are 2.68 and 4.10,

against 2.70 and 4.11 over all five starts; the abstract and conclusion quote the five-start values. Of SLSQP's 3.10-fold (S8) and 7.79-fold (S18) advantage over COBYQA, the gradient accounts for a factor of 1.71–2.47 and 2.68–4.10, and the optimizer for 1.22–1.76 and 1.91–2.92. The lower gradient factor and the higher optimizer factor belong to FD-warm, the cheaper finite-difference baseline; which baseline applies elsewhere depends on whether a solver can reuse the base point's state. COBYQA was slower than FD-warm from every start, by as little as 1.01 at S8 LHS 2, whereas against FD-cold it matched or beat SLSQP from two of five S8 starts (B/FD-cold 0.70 and 0.95). In seconds, which removes the 3% difference between the main and FD-warm sessions' units, B/FD-warm is 1.81 and about 3.0 (per start in Appendix D).

Table 5: The gradient factor and the optimizer factor at the primary targets: geometric means over the starts where B attains the target (five in S8, four in S18), each ratio formed within one session's units. The products close the main study's B/A only approximately, because their factors come from different sessions.

	S8, T_{90}	S18, T_{99}
Total: COBYQA against adjoint SLSQP (B/A, main study)	3.10	7.79
Against warm-started finite differences		
Gradient factor (FD-warm/A′)	1.71	2.68
Optimizer factor (B/FD-warm)	1.76	2.92
Product	3.02	7.82
Against cold-started finite differences		
Gradient factor (FD-cold/A′)	2.47	4.10
Optimizer factor (B/FD-cold)	1.22	1.91
Product	3.02	7.82

Where the time goes. Figure 7 splits each arm's mean time to the primary target into validation, the gradient and the rest (coarse forwards, mesh motion and optimizer work). The three SLSQP arms spend nearly the same time on validation and on the rest, because they cross after the same validated designs: together 5.3–5.4 units in S8 and 8.0–8.1 in S18. The gradient is what differs. A′'s adjoints take about 5% of its time to the target, 0.30 and 0.51 units, against 4.37 and 14.91 units (45% and 65%) for FD-warm's stencils and 8.72 and 27.47 units (62% and 77%) for FD-cold's. A free gradient would therefore shorten A′'s time to the target by only 5–6%: the end-to-end ratios are bounded by the validation and coarse forwards that every arm pays, not by the adjoint. This is why they fall well below the per-gradient cost ratios of 18–70 below.

Gradient cost. Costs are in each session's unit, the median cold validation forward (66.66 and 69.09 s). One adjoint gradient costs A′ 0.057 and 0.061 units with eight and 18 parameters. One finite-difference gradient costs FD-cold 1.90 and 4.25 units, about 0.24 units per cold stencil solve, 33 and 70 times the adjoint; it costs FD-warm 0.97 and 2.36 units, against 0.054 and 0.059 for the adjoint in FD-warm's session, 18 and 40 times as much. A warm stencil solve is charged about half a base coarse forward (Appendix D). Two factors shrink the per-gradient ratios. First, every coarse evaluation costs a coarse forward that both arms pay, 0.216–0.245 units (Table 4; Appendix D gives each session's medians), so an evaluation with a gradient costs FD-cold 7.4 and 14.7 times as much as A′. Second, A′ requests a gradient at every evaluation, whereas up to the primary crossing FD-cold and FD-warm request one at a median 83.3% of their evaluations in S8 and at 76.9% and 71.4% in S18. With both factors the expected ratios are 6.3 and 11.5 for FD-cold and 3.8 and 6.5

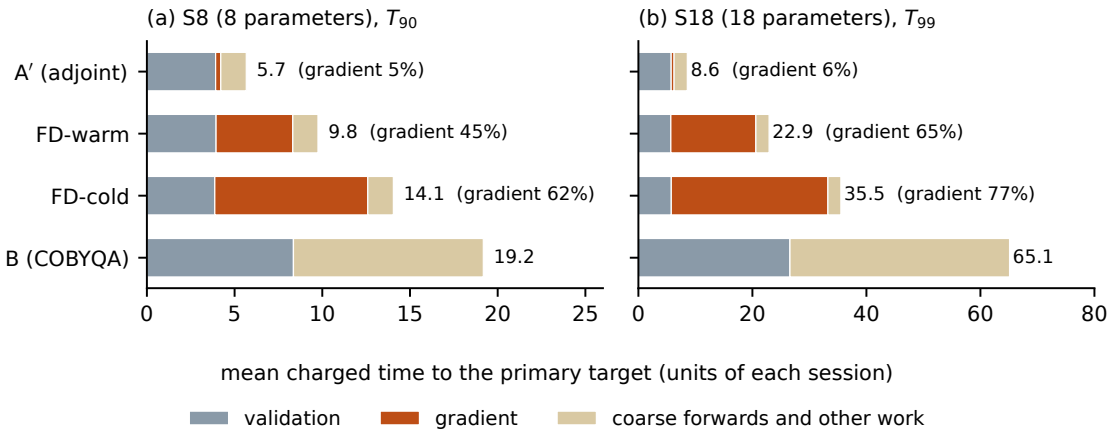


Figure 7: Mean charged time to the primary target, split into validation (all validations up to and including the crossing), the gradient (FD-warm's and FD-cold's stencil solves; A′'s adjoints, estimated as the number of evaluations times the run's mean adjoint charge per evaluation) and the rest, over the starts of Table 5. A′ is from the FD-warm session; FD-cold's own A′ differs by less than 1%. B has no gradient; its rest includes its initial interpolation set. Each bar is in its own session's unit; the bar labels give the total and, for the SLSQP arms, the gradient's share of it.

for FD-warm, against the measured 5.58 and 10.07 and 3.23 and 6.10 once validation is excluded; Appendix D discusses the 6–16% excess.

From 8 to 18 parameters (n grows 2.25-fold) the per-gradient cost ratio grows 2.2-fold against FD-warm and 2.1-fold against FD-cold, as a cost proportional to n predicts, but the end-to-end ratio grows only 1.6-fold (1.71 to 2.70) and 1.7-fold (2.47 to 4.11), because validation and coarse forwards do not grow with n. The two spaces also differ in primary target and in mesh-motion map; at the common target T_{90}, FD-warm/A′ is 1.71 in S8 and 2.59 in S18, and FD-cold/A′ 2.47 and 3.86. Two spaces do not determine a rate, and we do not extrapolate.

Gradient accuracy. FD-cold's L1 gradients pass the max-norm check, and their one component-wise failure is forward-difference truncation on a small component (Appendix D). At the frozen Newton tolerance, warm stencils are less accurate: FD-warm's max-norm errors are 0.92 to 1.08 times FD-cold's, but its component-wise check fails at all four points, on small components and once with a reversed sign, from an offset that the warm stencil objectives share and that a central difference removes (Appendix D). The final checks of the selected designs of all three arms keep every target crossing within its target at L3, with at least 15% margin (Appendix D).

7 What transfers, and limits

Three findings should transfer beyond this motor. First, SLSQP's advantage over a derivative-free search factors into a gradient factor and an optimizer factor, and both matter (Table 5). Second, with the optimization path held fixed, the gradient factor is a ratio of end-to-end times, not of gradient costs. The adjoint made the gradient about 5% of the time to target, so the time every arm spends on validation and coarse forwards bounds the factor: per-gradient cost ratios of 18–70 became end-to-end ratios of 1.71–4.11, and from 8 to 18 parameters the per-gradient ratio grew about as n while the end-to-end ratio grew 1.6- to 1.7-fold. The factor also depends on how the stencil is solved: warm-starting it halved the cost of a stencil solve and cut the gradient factor from 2.47 and 4.11 to 1.71 and 2.70, at some cost in accuracy on small components, and a cheaper

gradient still, such as a direct or semi-analytic sensitivity, would shift more of the advantage to the optimizer factor. Third, validating each incumbent compresses same-resolution ratios toward one and inflates SLSQP's lead over the coarse competitor D, which needs more validated incumbents (Section 5).

The evidence has clear limits. It is a paired case study over five deterministic starts of one motor; the ratios are not population estimates. The ablations are post hoc; FD-cold's forward differences carry a truncation error and FD-warm's warm stencils a Newton-tolerance offset, each exceeding the per-component check limit on small components (Section 6). CMA-ES and other population or multi-start methods remain untested. Method C ran with the preregistered settings of Appendix B and was not tuned; its low attainment bounds this configuration, not radial-basis surrogates in general. No reduced physical model, such as a nonlinear magnetic equivalent circuit (Kemmetmüller et al., 2015), a waveform-targeting surrogate (Farshbaf Roomi et al., 2022) or an adaptive reduced model (Wen and Zahr, 2023), was compared: D searches a coarse finite-element simulation, not such a baseline, and the slotless screening control of Appendix I cautions against inferring an advantage from parameter count alone. The forward model is two-dimensional and magnetostatic, without thermal, demagnetization, fatigue, end-effect or loss models, and is compared with published computational artifacts, not with measurements on a physical motor.

Cheap, verified gradients also make methods affordable that this study did not test. Second-order information is one: Hessian-vector products through a second-order adjoint, or quasi-Newton methods driven by exact gradients beyond the BFGS update inside SLSQP. Gradient-enhanced population and multistart searches are another, such as particle swarms or CMA-style searches seeded or steered by adjoint gradients, with runs in parallel across starts; every run here used one single-threaded core. Each still pays for validation and coarse forwards, which already bound the end-to-end gain (Fig. 7), and can gain beyond that only by needing fewer of them or, in wall time but not in compute, by running them in parallel. A natural next problem for the same pipeline is the rotor flux-barrier design of synchronous reluctance machines, a classic shape-optimization target because their torque ripple is high; gradient-based free-form shape optimization has already been applied to such a machine (Gangl et al., 2022).

8 Conclusion

On a common validation simulation, SLSQP with adjoint gradients reached the primary ripple targets 3.10 and 7.79 times sooner than COBYQA with 8 and 18 parameters, and post-hoc ablations split that advantage into two factors. The gradient factor, adjoint against forward differences in the same SLSQP, was 1.71 and 2.70 against warm-started stencils and 2.47 and 4.11 against cold ones; the optimizer factor, COBYQA against finite-difference SLSQP, made up the rest. The adjoint left the gradient about 5% of the time to target, so validation and coarse forwards now bound the remaining gain. Before trusting such a gradient, check what it differentiates: interface order and excitation rule change the objective or its sensitivities.

Author contributions

C.D. wrote the four-layer operator specification that GEODE realizes (Dunnavant, 2026), whose layers are unrelated to the mesh levels L0–L3, and gave design input and review. R.W. ran the work and the software agents that carried it out and is responsible for the results.

Code and data availability

The original forward and gradient evidence is fixed at GEODE commit `910593ba`. The repository contains the raw motor runs and analysis, geometry and mesh-generation scripts, design-gradient sweeps and timing records, and solver verification logs. Candidate-validation records are pinned at `224332c5`, and independent reference reruns at `8acd21ec`. Reference data are extracted from the authors' ICEM snapshot at `b145aaf4`, a later repository revision rather than a claim to the original conference release. Source hashes, extraction checks and the upstream GPL-3.0-or-later license are retained. Recomputing the physical study requires the separately documented mesh and solver commands; the benchmark READMEs record the GEODE-side Gmsh and Rust toolchain versions.

The matched optimization study (Section 5) follows the optimization harness at `9486a9da`, the source of the study's evaluator, and its coarse-gradient finite-difference records; the final-refinement binary was built from the program at `63b3e3d8`, which differs from the version at `3e4ee0c6` only by one unused import. This paper's PDF and LaTeX source are archived at doi:10.5281/zenodo.23580548. The paper package is the directory `papers/motor-gradient-optimization/motor-gradient-optimization.22` of the GEODE repository. It contains a manifest of every copied input with its source commit and SHA-256, the figure inputs and regeneration scripts (rebuilding the figures does not require rerunning FEM), four standard-library verifiers (one entry point that runs the other three) that recompute the reported numbers from those inputs, the per-run validation histories and ledger charges, the 213 final-check results, copies of the coarse-gradient checks, and the study's SHA-256 inventory of all 5,453 output files. The inventory was taken when the session finished; its existence by 2026-10-20 03:50 UTC is timestamped (RFC 3161 and OpenTimestamps), and each extracted run and session record is bound to it by hash. The session, calibration and anchor-recomputation records are included, as is an export of the governing preregistration comments taken on 2026-10-18 at 09:34 UTC, before the main session began at 18:16 UTC.

For the two post-hoc ablations (named `F` and `F_w` in their records), the package holds the 40 and 20 run extracts with their evaluation timelines, the setup records, each session's SHA-256 inventory (2,118 and 1,180 files) with its RFC 3161 and OpenTimestamps records, the final-check results, the comparison of the original and rebuilt refinement binaries, and an export of each protocol. FD-cold's evaluator source is `5cc86ad8`, its L1 gradient check is pinned at `74843242` and its step sweep at `f94acad3`; FD-warm's evaluator source and L1 checks are at `669029c5`. FD-cold's protocol was posted at 22:34 UTC on 2026-10-20 and frozen by a comment at 23:00 UTC, before the first FD-cold run began at 02:07 UTC on 2026-10-21; FD-warm's timeline is in Appendix D. These are GitHub timestamps, and neither protocol text was ever edited. The raw traces of all three sessions, the 5,453, 2,118 and 1,180 inventory-listed files with the inventories and their timestamp records, are packed in one archive with each session's final-check requests and results, the ablations' selection outputs and the binary comparison with the rebuilt binary: `motor-paper-raw-traces.tar.zst` (2,790,582,024 bytes, SHA-256 `1dfbf3883a9f1f3d8ddc44f67ce594b88291ec235a154337d24efa07a1b98b07`), deposited at DOI 10.5281/zenodo.23580550. Every file was checked against its inventory when the archive was built, and the archive's manifest lists the SHA-256 of every file it holds. Given the unpacked archive, the package's verifiers re-derive the run, timeline and final-check extracts of all three sessions from the raw traces and results.

A Field model details

A.1 Nonlinear field equation

The two-dimensional magnetic vector potential $A = A_z$ determines the flux density $\mathbf{B} = (\partial_y A, -\partial_x A)$. Iron is described by an isotropic reluctivity $\nu(|\mathbf{B}|^2)$, where reluctivity is the inverse of permeability. For a test function v that satisfies the homogeneous constraints, the weak equation is

$$\int_\Omega \nu(|\nabla A|^2)\nabla A \cdot \nabla v \,\mathrm{d}\Omega = \int_\Omega J_z v \,\mathrm{d}\Omega + \int_{\Omega_m} \mathbf{M} \cdot (\partial_y v, -\partial_x v)\,\mathrm{d}\Omega. \tag{4}$$

Here $\mathbf{M}$ denotes the permanent-magnet source in $\mathbf{H} = \nu\mathbf{B} - \mathbf{M}$. In the magnet, $\mu_r = 1.05$ and $B_r = 1.4\,\mathrm{T}$, so its source magnitude is $H_c = B_r/(\mu_0\mu_r)$, not B_r/μ_0. Its direction is $\pi/2$ in the rotor frame and rotates with the rotor. Air and windings have $\mu_r = 1$. The inner and outer circular boundaries have $A = 0$.

We use continuous piecewise-linear triangles. With nodal vector u, element gradients $g_i = \nabla\varphi_i$, $G = \nabla A$ and $s = G \cdot G$, the residual derivative on one element is

$$K^e_{ij} = |\Omega_e| \left[\nu(s) g_i \cdot g_j + 2\nu'(s)(g_i \cdot G)(g_j \cdot G)\right]. \tag{5}$$

The second term is required when the material changes with field strength. Newton steps use this consistent tangent, with damping and a stopping reference independent of the initial guess. Constant potential offsets, where allowed by the boundary conditions, do not change that reference. The solver reports whether it stopped at the requested tolerance or an estimated floating-point floor. All 500 baseline and sensitivity solves, and all 1,140 additional harmonic-coupling solves, stopped at the requested tolerance.

The iron law uses 47 tabulated M27 B–H points from 0 to 2.30 T. Monotonicity is imposed on $H(B)$, not on $\nu(B^2) = H(B)/B$. The tabulated H/B is non-monotone up to about 0.75 T. The two material tangent eigenvalues must satisfy

$$\nu > 0, \qquad \nu + 2s\nu' = \frac{\mathrm{d}H}{\mathrm{d}B} > 0. \tag{6}$$

The GEODE and reference interpolants agree to round-off between 0.05 and 2.25 T. GEODE uses an odd cubic near zero and a continuous slope into the vacuum-slope tail above 2.30 T. The reference uses a low-field clamp and a different last-knot slope. The maximum relative difference in reluctivity near zero is 6.66%; the last segment differs in H by at most 2.17×10^{-4} relatively. Appendix F tests the torque effect of the low-field difference.

A.2 Rotation, coupling and torque

At the modeled angles, the two circular traces have matching nodes after rotation. Our conforming interface identifies their potentials, including the antiperiodic sign on sector wrap-around. Writing the full nodal vector as $u = P\hat{u} + u_D$, the reduced residual and tangent are $\hat{R} = P^T R(P\hat{u} + u_D)$ and $\hat{K} = P^T K P$. The rotor geometry and magnet source rotate together; currents use the same mechanical angle in Eq. (1).

We average Maxwell stress over the air gap using the Arkkio expression (Arkkio, 1987)

$$T(\beta) = \frac{L}{\mu_0(r_2 - r_1)} \int_{\Omega_{\mathrm{gap}}} r B_r B_\theta \,\mathrm{d}\Omega, \qquad (r_1, r_2) = (44, 45)\ \mathrm{mm}. \tag{7}$$

Positive torque acts counter-clockwise on the rotor. All torques in this paper include axial length and refer to the modeled sector. We apply no factor of six when comparing with the saved reference arrays, despite a per-length label appearing in the reference evaluation plots.

The alternative harmonic model imposes

$$\int_{\Gamma_c} (A_s - A_r) e_n \, \mathrm{d}\Gamma = 0, \qquad e_n \in \{\sin(n\theta), \cos(n\theta)\}, \quad n = 3, 9, \ldots, N_h, \tag{8}$$

with maximum order $N_h = 45$ unless stated otherwise. The 16 order-45 constraints match the reference's eight orders. Rotor and stator retain distinct trace unknowns. With constraint matrix C, Lagrange multipliers η and reduced potential $\hat{u}$, Newton's method uses the saddle tangent

$$\begin{bmatrix} K & C \\ C^T & 0 \end{bmatrix}. \tag{9}$$

The trace moments use the exact Fourier integrals of the piecewise-linear hats. Rotation enters their centers; the constrained energy derivative gives

$$T_c = -L\eta^T \frac{\partial (C^T \hat{u})}{\partial \beta}, \tag{10}$$

with the state held fixed in the partial derivative, consistent with Egger et al. (2022). The studies pair couplings and estimators as stated in Section 3, with coupling torque as a separate check in the forward comparisons; the two objectives and their gradients are consequently comparisons of those specified model–estimator pairs.

The main harmonic runs retain antiperiodic trace-end pairing. An additional set frees those ends, matching the reference's treatment. The full-order identity check is summarized in Section 3.

A.3 Implementation checks

The nonlinear implementation is checked against linear assembly, residual finite differences, a manufactured solution and a saturated annulus with an independent radial field relation. Sector tests compare the constrained solution with a full-machine solve, check the reduced tangent by finite differences, and exercise positive, negative and wrapped rotation steps. Clockwise and counterclockwise triangle orderings give equivalent magnet sources and fields. The motor regression also detects changes in magnet strength, magnet direction, phase ordering and rotation direction. The verification records are linked in the data-availability section.

B Optimizer and search settings

Table 6 lists the settings a replication needs beyond Section 5; the pinned harness documents them in full. All methods work in the unit box $u \in [0, 1]^n$ of the free coordinates, minimize $F_r/F_r(\theta_0)$ on their coarse simulation r and impose the mean floor as $\bar{T}_{r,1}/\bar{T}_{r,1}(\theta_0) - 1.0025 \geq 0$. No method validates its start point, and a point validated once is not validated again.

Table 6: Optimizer settings. n is the number of free coordinates (8 or 18), x_v the last admissible evaluated point and x a refused request.

Method	Settings
A, E	SciPy SLSQP, `ftol` 10^{-15}, `maxiter` 10^5. Adjoint gradients of the objective and the mean constraint; analytic Jacobians of the geometry constraints. Safeguard: a refused request (unbuildable, mesh-infeasible or derivative undefined) aborts the current solve; the first admissible point $y_k = x_v + 2^{-k}(x - x_v)$, $k = 1, \ldots, 20$, is restored and SLSQP restarts from it with fresh state, at most 20 times. Validation: each iterate the SLSQP callback reports.
B, D	SciPy COBYQA, $2n + 1$ initial interpolation points, trust radius 0.1 to 10^{-6}, `maxiter` = `maxfev` = 10^9. A refused point returns NaN for objective and mean constraint; geometry constraints keep their values. Validation: the callback's best point whenever it changes, after the initial set.
C	Four cubic radial-basis interpolants, $\phi(r) = r^3$ with a linear tail: the three objective-current standard deviations, each over $\sqrt{F_r(\theta_0)}$, and the full-current mean, refitted after every evaluation. Initial design: the start and $u_0 \pm \Delta_0 e_i$, $\Delta_0 = 0.1$ ($n + 1$ points). Subproblem: SLSQP on the surrogate in the box and $\|u - u_k\|_\infty \le \Delta$, at most 200 iterations. Acceptance: a candidate better than the incumbent, feasible first (lower violation, then lower F). On acceptance with $\rho \ge 0.75$ and a step of at least 0.9Δ, $\Delta \leftarrow \min(2\Delta, 0.5)$; on rejection $\Delta \leftarrow \Delta/2$, with a geometry point $u_k \pm \Delta e_j$ after every second rejection. Stop at $\Delta < 10^{-6}$. Refused points are skipped and count as rejections. Validation: each acceptance.

The Latin-hypercube starts (McKay et al., 1979) use seeds 1–4; the computations use NumPy (Harris et al., 2020) and SciPy (Virtanen et al., 2020).

C Study protocol details

Coarse gradients. Section 4 verifies a different, scalar parameter set; the gradients that A and E use were checked separately. In both spaces, the L1 coarse objective F and full-current mean were compared with central differences at θ_0 and at an interior point, along a fixed box-scaled direction and for three (S8) or four (S18) components, including the offsets CPR1 and CPS1. With Newton tolerance 10^{-12}, the directional relative errors at $h = 10^{-4}$ are 1.5×10^{-8} to 2.8×10^{-7} for F and 2.4×10^{-9} to 4.7×10^{-9} for the mean, and the component errors at $h = 10^{-6}$ are at most 2.8×10^{-6}. The directional error falls by 72 to 156 times from $h = 10^{-3}$ to 10^{-4}, as truncation error should. The evaluator's own gradient replies at the coarse simulation's Newton tolerance of 10^{-8} agree at the interior point within 3.1×10^{-6} directionally; the largest component error, 4.3×10^{-5}, belongs to the small Sr1 derivative. At θ_0, S18's offsets sit on their bounds, so the central stencils there step outside the design box. The records are pinned with the optimization harness (Code and data availability).

Session. The session ran on one AWS c7a.4xlarge host (x86-64; NumPy 2.5.4 and SciPy 1.18.1 with OpenBLAS) with 15 concurrent single-threaded processes. Before calibration, the host recomputed the initial-design validation waveforms in both spaces. F_0 agreed with the canonical anchor to 2.4×10^{-13} relative, within the preregistered 10^{-9} tolerance, and the two spaces agreed bit for bit. The calibrated unit was 70.94 s, the median of 45 concurrent cold validations. All 100 runs completed in one session with no recovery, deferral or contamination: the one-minute load average peaked at 21.1, below the contamination threshold of $1.5N = 22.5$. The in-run cost totals 109 CPU-hours. Of the 100 runs, 38 stopped on reaching T_{99}, 46 on optimizer convergence and 13 at the budget. The

other three are SLSQP runs (A) in S8 whose solver stopped without converging: both repeats from LHS 2, after about 43 units, and repeat 1 from LHS 3, after 119 units.

Refinement host. The registered pipeline begins by capturing the closed study on its original host, a step that could not be completed after that host had been restarted. The refinement instead used a copy of the study outputs whose integrity is established by a SHA-256 inventory of every output file, taken when the session finished, and independently timestamped; the timestamps establish that the inventory existed by 2026-10-20 03:50 UTC. It ran on a different host (Apple M3 Ultra, macOS arm64). For the 16 selected designs validated on the moved reference mesh, the primary check reproduces the study's F_{val} to below 2.2×10^{-12} relative. The other 15 candidates needed a fresh mesh, which Gmsh triangulates differently on the two platforms; their F differs by at most 1.7×10^{-3} relative. The remaining two designs are θ_0 in each space, which reproduces F_0. The fresh-mesh differences measure the sensitivity of F to remeshing and are far below the 10% allowance δF.

D Per-start times, ratios and final checks

Table 7: Charged wall-time units to the first attaining validation, repeat 1. A dash means the target was not attained within 120 units. T_{99} was attained by no run in S8.

Space, target	Method	θ_0	LHS 1	LHS 2	LHS 3	LHS 4
S8, T_{50}	A	2.7	2.5	2.6	2.7	5.1
	B	8.5	9.3	9.3	8.9	13.7
	E	1.9	1.8	1.0	3.0	—
	D	3.1	5.3	7.3	4.8	8.8
	C	10.0	9.8	4.1	10.6	—
S8, T_{90}	A	2.7	2.5	7.4	6.4	8.8
	B	8.5	9.3	13.3	15.4	49.3
	E	1.9	1.8	4.5	3.0	—
	D	4.6	5.3	7.3	9.4	21.9
	C	16.9	9.8	—	—	—
S18, T_{50}	A	1.6	1.7	3.0	4.8	1.6
	B	12.2	13.1	13.5	21.5	15.7
	E	1.0	1.1	1.1	4.4	1.2
	D	4.1	6.9	5.2	11.9	8.4
	C	7.3	8.4	7.3	12.1	8.9
S18, T_{90}	A	3.1	5.4	3.0	12.0	1.6
	B	14.4	13.1	13.5	36.6	35.3
	E	2.0	—	2.8	10.9	1.2
	D	5.9	8.1	5.2	15.9	14.3
	C	8.4	8.4	10.3	—	14.6
S18, T_{99}	A	5.4	6.6	9.1	13.1	13.4
	B	90.4	59.1	48.6	—	62.3
	E	3.9	—	6.5	10.9	8.3
	D	43.0	14.1	36.0	83.3	47.8
	C	—	—	—	—	25.9

Table 8: Paired per-start time ratios (repeat 1). B/A and D/E vary the optimizer at fixed resolution; E/A and D/B vary the resolution; D/A compares A with the coarse derivative-free competitor. Ratios above one favor the denominator. Geometric means use the finite pairs only, so the D/E and E/A means omit the starts where E failed to attain the target.

Space, target	Ratio	Finite pairs	Geometric mean	Per-start ratios
S8, T_{50}	B/A	5	3.25	3.18, 3.66, 3.63, 3.25, 2.66
	D/E	4	2.75	1.60, 2.90, 7.68, 1.61, —
	E/A	4	0.68	0.72, 0.72, 0.37, 1.09, —
	D/B	5	0.56	0.36, 0.57, 0.78, 0.54, 0.64
	D/A	5	1.83	1.15, 2.09, 2.85, 1.75, 1.71
S8, T_{90}	B/A	5	3.10	3.18, 3.66, 1.80, 2.43, 5.61
	D/E	4	2.45	2.41, 2.90, 1.61, 3.17, —
	E/A	4	0.62	0.72, 0.72, 0.61, 0.47, —
	D/B	5	0.54	0.54, 0.57, 0.55, 0.61, 0.44
	D/A	5	1.68	1.72, 2.09, 0.99, 1.48, 2.49
S18, T_{50}	B/A	5	6.49	7.76, 7.70, 4.51, 4.46, 9.59
	D/E	5	4.69	3.90, 6.14, 4.93, 2.72, 7.09
	E/A	5	0.63	0.66, 0.66, 0.35, 0.91, 0.72
	D/B	5	0.46	0.33, 0.52, 0.39, 0.56, 0.53
	D/A	5	2.97	2.58, 4.04, 1.75, 2.48, 5.09
S18, T_{90}	B/A	5	5.06	4.64, 2.41, 4.51, 3.05, 21.54
	D/E	4	3.14	2.94, —, 1.86, 1.46, 12.17
	E/A	4	0.79	0.65, —, 0.94, 0.91, 0.72
	D/B	5	0.45	0.41, 0.62, 0.39, 0.44, 0.41
	D/A	5	2.25	1.91, 1.50, 1.75, 1.33, 8.74
S18, T_{99}	B/A	4	7.79	16.65, 8.92, 5.33, —, 4.65
	D/E	4	7.20	11.05, —, 5.50, 7.65, 5.79
	E/A	4	0.72	0.72, —, 0.72, 0.83, 0.61
	D/B	4	0.50	0.48, 0.24, 0.74, —, 0.77
	D/A	5	4.32	7.91, 2.12, 3.94, 6.36, 3.56

Table 9: Final checks of the selected designs. "best" marks a method's representative; "A T_{90}" marks the earliest repeat-1 crossing of that target by that method. The L2 column is the primary check on the refinement host. L3 and conforming changes are in percent; the 40-angle, tight and shifted changes are relative, in units of 10^{-4}. "held-out" is the objective at currents 0.25 and 0.75 relative to θ_0's at the same currents. A dash means the conforming check does not apply because stator offsets are nonzero.

Design	F_{L2}/F_0	L3 (%)	g40 (10^{-4})	tight (10^{-4})	shifted (10^{-4})	conf. (%)	held-out
S8							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.1	1.0000
best A	0.0432	−0.4	0.0	0.0	+0.1	+4.4	0.0472
best B	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best C	0.0489	−0.7	+0.2	0.0	+0.5	+4.1	0.0524
best D	0.0432	−0.5	0.0	0.0	0.0	+4.4	0.0472
best E	0.0434	−0.6	+0.4	0.0	+0.9	+4.4	0.0473
A T_{50}, A T_{90}	0.0653	−2.9	+0.7	0.0	+1.5	+1.1	0.0686
B T_{50}, B T_{90}	0.0881	−2.3	+1.3	0.0	+2.6	+3.2	0.0866
C T_{50}	0.1472	−4.0	+0.6	0.0	+1.3	+3.3	0.1508
C T_{90}	0.0507	−0.9	+0.2	0.0	+0.5	+4.0	0.0543
D T_{50}	0.0920	−2.2	+1.5	0.0	+3.0	+3.3	0.0899
D T_{90}	0.0594	−2.3	+1.3	0.0	+2.6	+1.6	0.0609
E T_{50}	0.1105	−2.0	+0.6	0.0	+1.1	+1.2	0.1104
E T_{90}	0.0660	−2.9	+0.7	0.0	+1.5	+1.1	0.0694
S18							
θ_0	1.0000	−1.3	0.0	0.0	+0.1	+2.0	1.0000
best A	0.0053	+2.6	+0.4	+0.1	+0.5	—	0.0066
best B, B T_{99}	0.0085	−3.0	−0.4	0.0	−1.2	—	0.0076
best C, C T_{99}	0.0057	−3.7	+2.5	0.0	+5.0	—	0.0074
best D	0.0044	+6.3	+2.6	+0.1	+4.9	—	0.0076
best E	0.0071	+2.6	−1.5	0.0	−3.1	—	0.0087
A T_{50}	0.1874	+0.9	−0.3	0.0	−0.6	—	0.1538
A T_{90}	0.0255	+1.1	−0.1	0.0	−0.2	—	0.0248
A T_{99}	0.0077	+0.4	−0.1	0.0	−0.3	—	0.0096
B T_{50}	0.1854	−1.5	−0.1	0.0	−0.3	—	0.1907
B T_{90}	0.0523	−0.4	−0.2	0.0	−0.4	—	0.0505
C T_{50}	0.3958	−1.4	−0.1	0.0	−0.1	—	0.3692
C T_{90}	0.0259	+0.7	0.0	0.0	0.0	—	0.0247
D T_{50}	0.1925	−1.5	−0.1	0.0	−0.3	—	0.1971
D T_{90}	0.0393	−0.7	+0.6	0.0	+1.2	—	0.0392
D T_{99}	0.0089	+0.4	−0.5	0.0	−1.0	—	0.0137
E T_{50}	0.1961	+0.8	0.0	0.0	0.0	—	0.1567
E T_{90}	0.0223	−2.8	−5.3	0.0	−10.6	—	0.0144
E T_{99}	0.0081	+0.7	+0.2	0.0	+0.4	—	0.0112

Validation and the A–D contrast. Excluding the validation charges up to each crossing (Section 5.2), two of the ten S8 D/A pairs favor D (0.97 and 0.54). We conjecture that part of the larger S18 optimizer-contrast ratios comes from COBYQA's initial interpolation set, which costs $2n + 1 = 37$ coarse evaluations before its first model step there, against 17 in S8.

Held-out currents. The held-out currents test the ripple reduction outside the three objective currents. We compare each design's objective at currents 0.25 and 0.75, relative to θ_0's at the same currents, with its relative objective at the objective currents. Across the candidates, the held-out ratio is 0.65 to 1.73 times the objective-current ratio; values above one mean a smaller reduction at the held-out currents. For the five best S8 designs it is 1.07–1.09. In S18, four of the five best designs keep less of their reduction: their remaining objective fraction at the held-out currents is 1.23–1.73 times that at the objective currents, while for B's best design it is 0.89. D's best design has $0.0044F_0$ at the objective currents but 0.0076 of θ_0's value at the held-out currents. Part of the extra reduction from S18's added freedom is therefore specific to the three objective currents.

Table 10: Validation charges, repeat 1. "To crossing" counts the validations up to and including the first attaining one, and "share" gives the fraction of the time to that target charged to validation, as medians (ranges) over the method's attained targets and starts. "Run" gives the range over starts of the fraction of the whole run's charged time spent in validation.

	S8			S18		
Method	To crossing	Share	Run	To crossing	Share	Run
A	2 (2–7)	0.65 (0.65–0.71)	0.45–0.71	3 (1–10)	0.61 (0.58–0.68)	0.65–0.68
B	5 (2–25)	0.46 (0.22–0.53)	0.24–0.34	8 (2–36)	0.34 (0.15–0.45)	0.35–0.49
C	6.5 (2–10)	0.53 (0.42–0.64)	0.02–0.63	4 (2–17)	0.40 (0.25–0.56)	0.50–0.61
D	6 (2–20)	0.81 (0.60–0.84)	0.41–0.64	9 (2–70)	0.71 (0.46–0.77)	0.73–0.77
E	2 (1–5)	0.90 (0.87–0.92)	0.82–0.90	3 (1–11)	0.88 (0.83–0.90)	0.87–0.90

Ablation gradients and the optimizer factor. On the L1 coarse simulation with the frozen step, FD-cold's gradients agree with the adjoint within a max-norm relative error of 5.2×10^{-4} to 1.9×10^{-3} for F and at most 7.2×10^{-4} for the full-current mean at θ_0 and an interior point in each space, but one small component exceeds the check's per-component limit of 10^{-1} through forward-difference truncation, which a central difference with the same step reduces from 0.24 to 2.2×10^{-4} relative (below). In the FD-warm session a warm stencil evaluation takes a median of 60 Newton iterations over its 60 field solves, against 170 and 167 for a cold coarse evaluation, and is charged a median of 0.117 and 0.126 units against 0.220 and 0.233 for the session's base coarse forwards, about half. B/FD-warm per start, in each session's units, is 1.83, 2.07, 1.01, 1.38 and 3.23 in S8 (geometric mean 1.76) and 6.11, 3.35, 1.94 and 1.82 in S18 (2.92); in seconds, which removes the 3% difference in time unit, the means are 1.81 and about 3.0.

Cost estimate and measurement. The two-factor estimates of Section 6 run 6–16% above the measured ratios. Their cost model counts only coarse forwards, adjoints and stencil solves, so it omits the per-evaluation overhead of both arms (evaluator start-up, mesh motion and optimizer work), and it uses whole-run medians where the measurement runs to the crossing. The coarse forward that every coarse evaluation costs is taken per session: as a median over runs of coarse-simulation time per coarse forward, it is 0.240 and 0.236 units for FD-cold against 0.232 and 0.245 for its A′, and 0.220 and 0.238 for FD-warm's A′. FD-warm's coarse-simulation time also contains its warm stencils, so for FD-warm we take the median charge of one base coarse forward, 0.220 and 0.233 units (0.216 and 0.233 for its A′ on the same measure). Up to the crossings, an A′ evaluation costs a median 0.32 to 0.34 units, overhead included, against the cost model's 0.27 to 0.31, an excess of 10–17%. The finite-difference arms' evaluations differ from their cost model by +7% for both FD-cold and FD-warm in S8, and by −5% (FD-cold) and +1% (FD-warm) in S18. The overhead weighs most against A′'s cheap evaluations, which is where the cost model errs most. The gradient shares up to each crossing are counted from the evaluation timelines in the package.

Validation in the ablation. Validation is a median 61–65% of A′'s time to a target and 16–27% of FD-cold's (Table 12), which pulls the measured ratio toward one.

Ablation runs after the crossings. After the crossings in S8, FD-cold's SLSQP met its stopping test in two of ten runs and ran to the budget in the other eight, whereas A′ converged in four, stopped in the solver in four and reached the budget in two. The extra time bought FD-cold nothing: from every start its best validated objective is within 0.4% of A′'s. Why FD-cold rarely met the

stopping test (`ftol` 10^{-15}, Table 6) was not tested; it does not affect the time to target, since every crossing happens at the same validation count.

Control. As a check of the control, A′'s primary crossing times in the FD-cold session are 0.94–1.04 times those of the main-study A from the same start and repeat (0.97–1.05 in the FD-warm session), each in its own session's unit.

Coarse and validation objectives. The coarse simulation also reads lower than validation. At θ_0, the order-45 coarse objective is 0.887 (L1) and 0.894 (L0) of F_0. At the 31 selected designs, the ratio of coarse to validation objective lies between 0.57 and 1.19 times its θ_0 value. The lowest values belong to S18 designs near T_{99}: E's and A's T_{99} crossings (0.57 and 0.71) and D's best design (0.65); the best S18 designs of B and C are at 0.99 and 1.19. Large reductions therefore often, but not always, come with a coarse objective that underestimates validation more than at θ_0. This is consistent with the cancellation conjecture of Section E.2 but does not establish it.

Ablation session and finite-difference accuracy. On the L0 coarse simulation, FD-cold's gradients agree with the adjoint to a max-norm relative error of 5.1×10^{-4} to 1.9×10^{-3} for F and 1.0×10^{-4} to 6.7×10^{-4} for the full-current mean, at θ_0 and an interior point in each space. Single small components are less accurate, up to 5.7×10^{-2} for SW3 at the S18 interior point. These errors, at the frozen step, are part of what FD-cold is; the frozen acceptance limits were 5×10^{-3} for the norm and 10^{-1} per component. The last gradient of each S8 θ_0 run of FD-cold was cut off by the budget and is excluded from the cost medians. The L1 check was run after the ablation, with the frozen step and the frozen limits; it was not part of the ablation's protocol. Its status is FAIL because of one component: at the S8 interior point, the SW3 component of the mean gradient is -9.62×10^{-3} by finite differences against -7.74×10^{-3} by the adjoint, a relative error of 0.24, on a component 8.7×10^{-5} times the largest. The records show a pattern rather than scatter. For every one of the eight scalar components, the finite-difference-minus-adjoint error of the mean gradient has the same sign at all four points, at L1 and at L0, and for SW3 it is nearly the same at all four (three distinct designs, since θ_0 is shared by both spaces): -1.8 to -1.9×10^{-3} at L1 and -3.0 to -3.1×10^{-3} at L0. The SW3 failure is therefore this offset on a component whose adjoint value is small at that design. A step sweep at the failing point, recorded separately from the L1 check, identifies the offset. For SW3, MH and SW2, forward differences at steps of 10^{-5}, 10^{-6} and 10^{-7} m give an error that, divided by the step, keeps its sign and its size to within 0.9% (3.0% for the MH component of the objective gradient), as truncation error does and noise would not. Central differences remove it: the SW3 mean component's error falls from 0.24 relative (forward, 10^{-6}) to 2.2×10^{-4} (central, 10^{-6}), and its central error falls 94-fold from 10^{-5} to 10^{-6}, close to the 100-fold that h^2 predicts. At 10^{-6} the remaining central errors of the MH and SW2 components are 9×10^{-8} to 3.4×10^{-7} in absolute terms, so computational noise at that step is of that order or smaller, unless it cancels against the remaining truncation term. For the mean gradient, the magnitude of SW3's forward error per unit step, 1.9×10^3 N m/m^2, is smaller than MH's (2.8×10^3 N m/m^2) and SW2's (9.7×10^3 N m/m^2), so SW3 fails because its mean-gradient slope at this design is small (-7.7×10^{-3} N m/m), not because its curvature is large. In the terms of Moré and Wild (2011, 2012), the frozen step lies where truncation, not computational noise, dominates the forward-difference error. The sweep is deterministic: a repeated base solve agrees bit for bit, and its forward values at 10^{-6} equal those of the L1 check.

The session ran on the same type of host as the main study (c7a.4xlarge, 15 concurrent processes) with a calibrated unit of 66.66 s, against 70.94 s in the main session. The anchor recomputation

again agreed within 2.4×10^{-13}, no run was host-contaminated, and the one-minute load peaked at 19.7. A SHA-256 inventory of the 2,118 session files was taken on the host before it stopped and timestamped by two RFC 3161 authorities; every ablation number in this paper is recomputed from traces bound to it.

Ablation final checks. The final checks confirm FD-cold's and A′'s selected designs (Table 13). All 90 checks of the 14 designs completed. The primary check reproduces the study's F_{val} within 6.3×10^{-4} relative. L3 changes F by -2.9% to $+2.6\%$ (-2.9% to $+1.1\%$ at the crossings), the 40-angle, tight and shifted checks by less than 0.03%, and the conforming estimator gives 1.1% to 4.4% more ripple on the S8 designs. All ten target crossings keep their target at L3 with at least 15% margin, and every candidate keeps the mean floor, the closest by 0.044%. An independent selection program chose the same 14 designs, bit for bit.

Table 11: Post-hoc ablation (Section 6): charged time units to the first attaining validation for A′ and FD-cold, repeat 1, and their ratio per start; the last column gives the geometric mean, with repeat 2 in parentheses. Units are those of the ablation session.

Space, target		θ_0	LHS 1	LHS 2	LHS 3	LHS 4	GM (rep. 2)
S8, T_{50}	A′	2.8	2.6	2.7	2.8	5.2	
	FD-cold	6.9	6.5	6.5	6.7	12.7	
	FD-cold/A′	2.45	2.45	2.39	2.43	2.43	2.43 (2.44)
S8, T_{90}	A′	2.8	2.6	7.6	6.4	8.9	
	FD-cold	6.9	6.5	18.9	16.3	21.8	
	FD-cold/A′	2.45	2.45	2.50	2.52	2.45	2.47 (2.49)
S18, T_{50}	A′	1.6	1.6	3.2	4.8	1.7	
	FD-cold	6.3	5.6	11.3	17.8	5.9	
	FD-cold/A′	3.91	3.59	3.58	3.69	3.60	3.67 (3.68)
S18, T_{90}	A′	3.1	5.1	3.2	12.1	1.7	
	FD-cold	12.2	20.8	11.3	49.9	5.9	
	FD-cold/A′	3.92	4.09	3.58	4.14	3.60	3.86 (3.84)
S18, T_{99}	A′	5.4	6.3	9.3	13.2	13.6	
	FD-cold	23.6	24.0	38.7	54.9	55.8	
	FD-cold/A′	4.33	3.84	4.14	4.17	4.10	4.11 (4.04)

Table 12: Validation charges in the ablation, repeat 1. "To crossing" counts the validations up to and including the first attaining one; "share" is the fraction of the time to that target charged to validation, as medians (ranges) over all attained targets and starts; "run" is the range over starts of the fraction of the whole run's charged time spent in validation.

	S8			S18		
Method	To crossing	Share	Run	To crossing	Share	Run
A′	2 (2–7)	0.65 (0.65–0.70)	0.44–0.70	3 (1–10)	0.61 (0.58–0.68)	0.64–0.68
FD-cold	2 (2–7)	0.27 (0.26–0.29)	0.25–0.28	3 (1–10)	0.16 (0.15–0.18)	0.15–0.18

Table 13: Final checks of the ablation's selected designs, as in Table 9: L2 objective, L3 change, the largest 40-angle, tight or shifted change, and the conforming change where stator offsets are zero.

Design	F_{L2}/F_0	L3 (%)	max g40/tight/shifted (10^{-4})	conf. (%)
S8				
θ_0	1.0000	−1.3	0.1	+2.1
A′ T_{50}, T_{90}	0.0653	−2.9	1.5	+1.1
FD-cold T_{50}, T_{90}	0.0653	−2.9	1.5	+1.1
best A′	0.0432	−0.4	0.1	+4.4
best FD-cold	0.0432	−0.6	0.1	+4.2
S18				
θ_0	1.0000	−1.3	0.1	+2.0
A′ T_{50}	0.2721	−0.6	3.0	—
A′ T_{90}	0.0255	+1.1	0.2	—
A′ T_{99}	0.0077	+0.4	0.3	—
FD-cold T_{50}	0.2720	−0.6	3.0	—
FD-cold T_{90}	0.0252	+1.0	0.6	—
FD-cold T_{99}	0.0076	+0.4	0.3	—
best A′	0.0053	+2.6	0.5	—
best FD-cold	0.0054	+2.5	0.2	—

Paths before the crossings. Up to the primary crossing the validated objectives of FD-cold and A′ agree within 0.4% in S8 and within 0.9% at three S18 starts; at S18 LHS 2 and LHS 4 the iterates drift apart before the T_{99} crossing, by 29% at the crossing validation and 18% one validation before it, respectively, and at LHS 2 FD-cold uses one coarse evaluation fewer, yet each crossing comes at the same validation. FD-warm and its A′ cross every target after the same number of validations as FD-cold from every start, except FD-warm's T_{50} crossing at S18 LHS 3: its third validated objective, $0.46F_0$, misses T_{50} once the allowance is added, and its next five fall below the mean floor, so it crosses six validations later (the per-start ratio of 6.92 below).

Warm-stencil arm FD-warm. Its protocol was posted at 15:18 UTC on 2026-10-21 and frozen by a comment at 17:54 UTC, before the first FD-warm run began at 20:30 UTC (GitHub timestamps; the protocol text was never edited). The freeze comment recorded the result of the L1 accuracy gate and fixed the warm-start source. The session ran on the same host type with 15 concurrent processes and a calibrated unit of 69.09 s; the anchor recomputation agreed within 2.4×10^{-13}, no run was host-contaminated, and the one-minute load peaked at 17.4. A SHA-256 inventory of its 1,180 files was taken on the host before it stopped and timestamped by two RFC 3161 authorities. In S8, FD-warm ran to the budget in four of five runs and converged in one, and its A′ converged in two, stopped in the solver in two and reached the budget in one; in S18 all ten runs stopped at T_{99}. The other targets give FD-warm/A′ of 1.69 in S8 and 2.59 (T_{90}) and 3.02 (T_{50}, with one start at 6.92) in S18 (Table 14). Of FD-warm's 263 gradients, 262 warm-started every stencil from its base; one, at 110 units in S8 LHS 2, long after that run's crossings, fell back to cold stencils because its base had not been retained. This session's A′ reproduces the FD-cold session's A′ at the primary targets within 4% in each session's units, and the selected A′ designs coincide for six of seven roles; the earliest S18 T_{50} crossing differs because three starts cross within 1.56–1.60 units of each other and the timing noise reorders them.

On the L1 check, FD-warm's max-norm relative errors are 0.92 to 1.08 times FD-cold's at the four points, which passes the frozen gate of ten times FD-cold's, and its component-wise check fails at every point: SW3 of the mean gradient at the S8 interior point (as for FD-cold), Sr1 of

the objective gradient at both θ_0 points, and SW1, SW4 and Sr1 at the S18 interior point. In the step sweep at that point the forward errors of the three failing objective-gradient components are -3.1×10^{-4} to -3.8×10^{-4}, -3.1×10^{-3} and -3.0×10^{-2} to -3.1×10^{-2} at steps of 10^{-5}, 10^{-6} and 10^{-7} m, and central differences at 10^{-6} bring them below 1.2×10^{-7}. In the sessions, 58 of S8's 1,810 warm stencil evaluations and 20 of S18's 648, 3.2% in all, recorded fewer than 60 Newton iterations, as few as 40, so at least one of their 60 field solves took no Newton update and returned the base state. They perturb SW1, SW4 or Sr1 in S8 and SW4 or Sr1 in S18, all among the objective-gradient components that fail the L1 check. On a single thread of the check host, a warm stencil takes 0.32–0.37 of a cold stencil's Newton iterations and 0.48–0.53 of its wall time.

In detail, three of the four failures are on small objective-gradient components (SW1, SW4 and Sr1, 9×10^{-5} to 6.3×10^{-4} of the largest), with a sign reversal on Sr1 at the S18 interior point (relative error 1.58), and the fourth is at the S8 interior point on SW3 of the mean gradient, where truncation dominates at the frozen step as for FD-cold (a forward error per unit step of $-1.7 \times 10^{3}\,\mathrm{N\,m/m^2}$, against FD-cold's -1.9×10^{3}); unlike FD-cold's, this error changes sign at 10^{-7} m. At the S18 interior point the three failing components share one signed forward error, $-3.11 \times 10^{-3}\,\mathrm{N^2\,m^2/m}$ at a step of 10^{-6} m to within 0.3%, which grows about tenfold per decade of decreasing step and which a central difference removes. The stencil objectives there thus differ from the base value by a common offset of about $-3 \times 10^{-9}\,\mathrm{N^2\,m^2}$, whichever component is perturbed; we did not isolate its source.

The final checks of FD-warm's and its A′'s selected designs completed all 84 checks of 13 designs (Table 13 has the corresponding FD-cold results). The primary check reproduces the study's F_{val} within 4.1×10^{-4} relative; L3 changes F by -2.9% to $+2.6\%$ (-2.9% to $+1.1\%$ at the crossings), the 40-angle, tight and shifted checks by at most 0.03%, and the conforming estimator gives 0.9% to 4.4% more ripple on the S8 designs. All ten target crossings keep their target at L3 with at least 15% margin, every candidate keeps the mean floor, the closest by 0.044%, and an independent selection program chose the same 13 designs, bit for bit. These checks used a rebuilt refinement binary: the pinned one (SHA-256 `679a40c8`) was deleted with its build directory and rebuilt from the same source commit, `63b3e3d8`, as `c75fe847`. Twelve earlier checks covering all seven check types, rerun with the rebuilt binary, reproduce every non-timing result field (4,181 fields) exactly.

Table 14: Warm-stencil arm: charged time units to the first attaining validation for A′ and FD-warm (repeat 1, FD-warm session units) and their ratio per start, with the geometric mean.

Space, target		θ_0	LHS 1	LHS 2	LHS 3	LHS 4	GM
S8, T_{50}	A′	2.8	2.7	2.7	2.7	5.3	
	FD-warm	4.7	4.5	4.6	4.6	9.0	
	FD-warm/A′	1.68	1.68	1.70	1.70	1.69	1.69
S8, T_{90}	A′	2.8	2.7	7.5	6.3	9.0	
	FD-warm	4.7	4.5	13.1	11.2	15.3	
	FD-warm/A′	1.68	1.68	1.75	1.77	1.69	1.71
S18, T_{50}	A′	1.6	1.6	3.1	4.8	1.6	
	FD-warm	3.9	3.9	7.6	32.9	3.9	
	FD-warm/A′	2.50	2.41	2.42	6.92	2.50	3.02
S18, T_{90}	A′	3.1	5.3	3.1	11.8	1.6	
	FD-warm	7.7	14.4	7.6	32.9	3.9	
	FD-warm/A′	2.51	2.74	2.42	2.79	2.50	2.59
S18, T_{99}	A′	5.5	6.5	9.3	13.0	13.1	
	FD-warm	14.8	17.6	25.1	36.2	34.2	
	FD-warm/A′	2.69	2.73	2.69	2.79	2.62	2.70

E Forward reproduction details

Each mesh level contains 60 baseline solves: three current levels and 20 angles. For a scalar statistic q, the estimated L3 refinement interval is

$$q_3 \ \pm \ e_q, \qquad e_q = \max\{|q_2 - q_1|, |q_3 - q_2|\}. \tag{11}$$

We use the same construction pointwise for waveforms. This interval is a refinement diagnostic, not a proven error bound or a statistical confidence interval. It does not cover reference-model error or angular quadrature error. The loaded means change direction between the last two increments, so a last-increment estimate alone would be particularly weak.

Figure 9 in Appendix F shows the improvement angle by angle. The largest full-current difference falls from 0.0272 to 0.0034 N m. The harmonic full-current ripple interval contains the saved reference; the saved half-current value lies approximately 5×10^{-5} N m below its interval. Mean agreement is therefore insufficient to characterize waveform accuracy.

The saved zero-current reference exposes a separate discretization effect. Harmonic coupling changes the cogging standard deviation from 29.1% below to 12.9% above the reference. Its waveform is close to a 1.13-fold scaling of the saved waveform, but the reference lies outside the estimated ripple interval. This motivates refining the reference itself before interpreting the discrepancy. The harmonic cogging standard deviations are 0.0101, 0.0119, 0.0121 and 0.0121 N m on L0–L3, compared with 0.0107 N m in the reference. The independent reference refinement below tests this coarse-reference explanation directly.

Two additional controls narrow the explanation. At L3, coupling torque and the inner- and outer-half Arkkio estimates are each within 9×10^{-5} N m of full-band Arkkio in the order-45 solution. Freeing the coupling-arc ends changes torque by less than 5.4×10^{-5} N m. These effects, and the material, tolerance and winding-area controls of Appendix F, are too small to explain the remaining cogging gap at the tested resolution.

E.1 Independent reference refinement

We rerun the public reference with its own degree-two isogeometric model, harmonic orders $3, 9, \ldots, 45$ and coupling torque, at 3, 6, 12 and 24 subdivisions in each patch direction. The 3×3 rerun first reproduces all 60 saved torques with differences at most $9.8 \times 10^{-13}\,\mathrm{N\,m}$. The run uses GNU Octave 11.3.0 (Eaton et al., 2026), GeoPDEs (de Falco et al., 2011; Vázquez, 2016) at revision `53ae5ca1` and nurbs 1.4.3; compatibility edits, source hashes and raw records are retained. The corresponding field degrees of freedom are 4,079, 12,518, 43,220 and 159,920.

The reference cogging standard deviation changes from 0.01071 to 0.01220, 0.01212 and $0.01208\,\mathrm{N\,m}$ (Fig. 8a). GEODE L3 at order 45 gives $0.01209\,\mathrm{N\,m}$, a 0.10% difference from the finest reference. The maximum pointwise discrepancy drops from 2.1×10^{-3} to $3.4 \times 10^{-5}\,\mathrm{N\,m}$. At half and full current, the finest reference standard deviations are 0.03947 and $0.08014\,\mathrm{N\,m}$; GEODE is 0.35% and 0.27% higher (panel b). All five finest-reference statistics, the three standard deviations and two loaded means, lie within GEODE's L3 refinement intervals. The coarse saved reference therefore accounts for the cogging mismatch in Section 3.4 at the matched interface order.

This agreement still concerns finite discretizations. The last cogging increments approximately halve; extending that trend would put a limit near $0.0120\,\mathrm{N\,m}$, an estimate without a bound. The last two half-current standard-deviation increments are both about -0.3% and do not shrink; we therefore leave its remaining error unestimated. The 0.10% cogging agreement is a comparison of finite-resolution values, not an accuracy bound; neither refinement sequence is fully converged. Tightening the reference Newton stopping threshold from 10^{-6} to 10^{-14} changes the checked cogging torques by at most $4.4 \times 10^{-8}\,\mathrm{N\,m}$. GEODE retains straight-sided geometry, and the high-field extrapolation of the iron law has not been varied in a paired torque experiment. Neither cross-code agreement nor refinement proves physical accuracy for the machine.

E.2 The saved optimized design

The saved reference design at iteration 120, denoted Opt120, includes 12 scalar parameters and ten grouped surface-control offsets. We evaluate its full exported geometry on freshly generated meshes; simply moving the finest initial mesh produces four inverted triangles. Its inner air-gap band no longer has the annular area assumed by the full-band Arkkio functional, so harmonic results here use coupling torque. Conforming results use only outer-band Arkkio on the fresh exported mesh. On the same L3, order-45 Opt120 field at full current, the outer-band and coupling estimates differ by about $1.3 \times 10^{-4}\,\mathrm{N\,m}$ in mean and $1.1 \times 10^{-5}\,\mathrm{N\,m}$ in standard deviation. This paired check supports the estimator substitution at this resolution. This estimator restriction is specific to these geometries; it does not validate outer-band Arkkio on arbitrary moved meshes or stator offsets.

With the evaluation current held fixed in the unscaled geometry, raising the transmitted order from 45 to 93 multiplies Opt120's F by 3.6, from 1.07×10^{-6} to $3.86 \times 10^{-6}\,\mathrm{N^2\,m^2}$ (Fig. 8c). The initial design's F rises by 16%. Its ratio to Opt120 is consequently 7,631 at order 45, 2,454 at order 93 and 2,444 for conforming coupling. The full-current Opt120 standard deviation rises from 0.00066 to $0.00164\,\mathrm{N\,m}$. The large ripple reduction survives these checks, but its magnitude depends strongly on the interface model: the order-93 change is about 96 times the last spatial-refinement increment of F (Appendix G).

A possible explanation is that optimization against order 45 favors cancellation within that truncated model. This is a conjecture, not an ablation result: increasing the order raises full-current ripple but lowers Opt120 cogging standard deviation from 0.00061 to $0.00035\,\mathrm{N\,m}$. Appendix D reports the corresponding coarse-to-validation ratios at the designs selected in the optimization study. Opt120 is also outside the fixed-size, fixed-physical-current problem of Section 5: it shortens the stack from 100 to 80 mm and scales the radius by 1.0477, so its mean torque must be compared

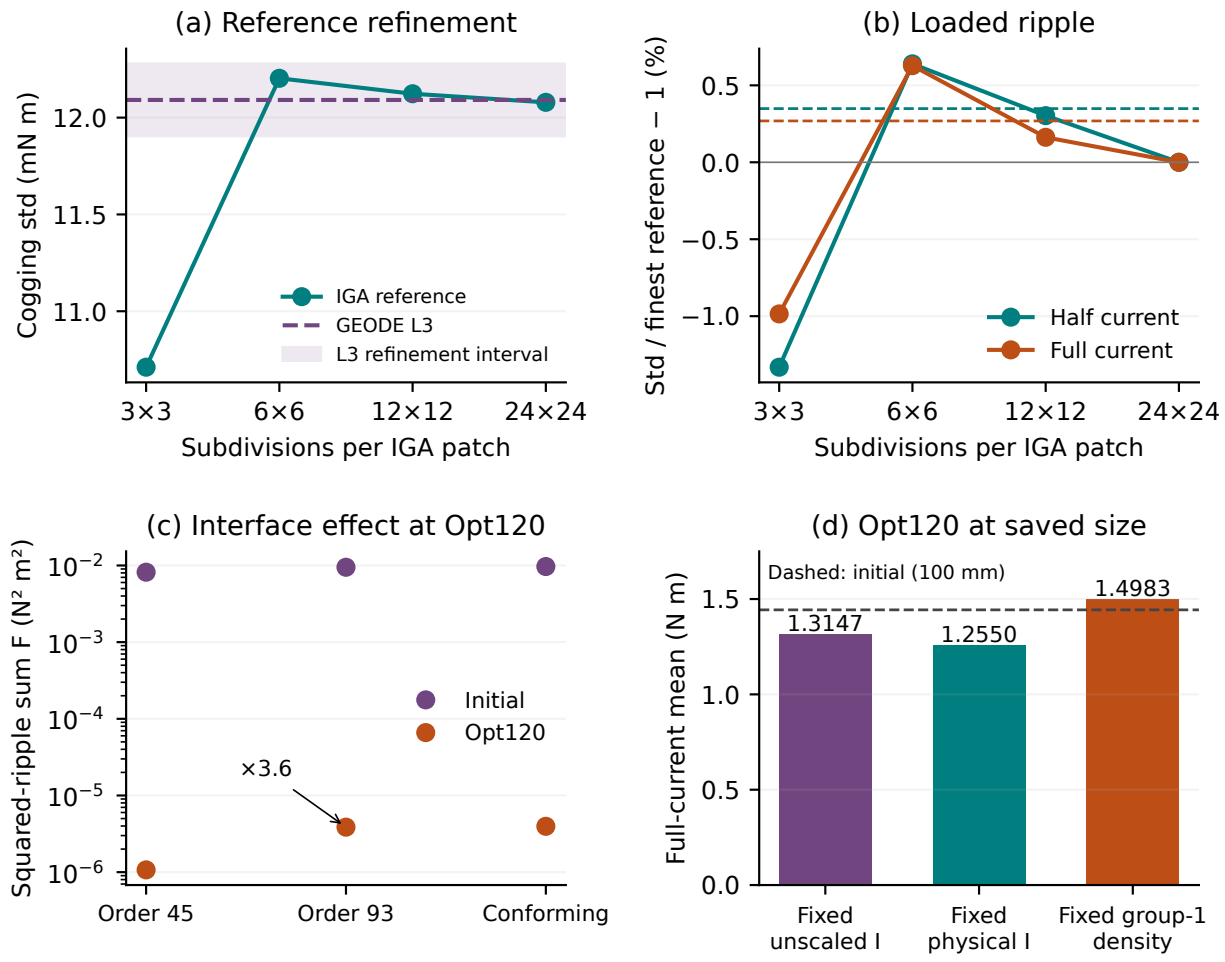


Figure 8: Reference and candidate validation from raw torque records. (a) Independent IGA reference cogging refinement at order 45; the dashed line is GEODE L3 and its shaded band uses Eq. (11). (b) Loaded standard deviation relative to the finest reference at each current; circles are IGA, horizontal lines are GEODE L3. The finest IGA point is zero by construction. Panels (a,b) use the initial geometry and 20 angles; GEODE uses Arkkio, with coupling torque checked separately. (c) Initial and Opt120 squared ripple sums at L3 on fresh meshes and 20 angles, under fixed unscaled current. Harmonic values use coupling torque; conforming values use outer-band Arkkio. (d) Full-current mean at order 45 for Opt120 under three current rules at its saved size (Appendix G); the dashed line is the initial mean at the initial size. All candidate values use coupling torque.

with the initial motor only after both size factors and the current rule are fixed. Appendix G gives those normalizations, a reproduction comparison at the saved reference current, and the effect of angular sampling at this low ripple.

F Forward-model controls

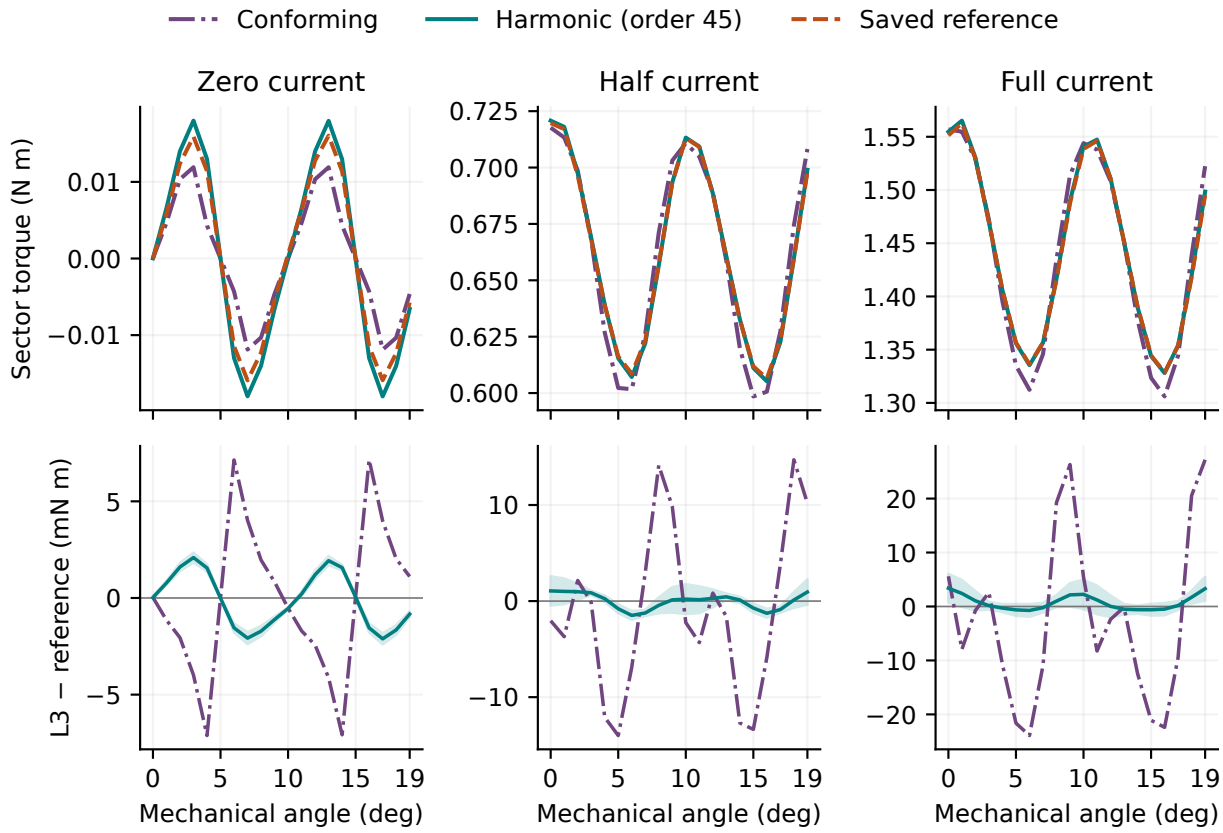


Figure 9: Top: L3 torque for conforming and order-45 harmonic coupling, compared with the saved reference. Bottom: each GEODE waveform minus the reference; shading is the harmonic model's pointwise estimated refinement interval from Eq. (11). Each residual panel has its own vertical scale, so compare residual magnitudes by their axis values, not by curve height. All GEODE curves use Arkkio torque, so the paired difference changes the coupling while holding the estimator fixed. The saved reference is coarse; Section E.1 reports its refinement.

The controlled changes in Table 15 are too small to explain the observed waveform gap at the tested settings. The near-zero material difference has particularly little effect on torque despite its 6.66% local reluctivity difference. This is a measured sensitivity of the initial design and does not establish insensitivity after optimization.

Table 15: Maximum absolute torque change relative to the baseline at the same mesh level, using conforming coupling and Arkkio torque. Entries summarize the recorded angle/current subsets.

Change	Levels	Max. $\lvert\Delta T\rvert$ (N m)
Reference low-field interpolation and clamp	L2, L3	2.1×10^{-6}
Relative Newton tolerance 10^{-12} vs 10^{-8}	L2; L3 full current	7.3×10^{-8}
Exact winding areas vs mesh areas	L2	3.9×10^{-4}
Inner or outer half of the Arkkio band	L3	7.3×10^{-5}

In the recorded estimator study, single-circle Maxwell-stress estimates vary more with the

integration radius than the half-band Arkkio estimates. The paired coupling-torque and half-band checks of Section E avoid selecting a favorable radius. A post-processing filter of the solved conforming field also gives non-monotone ripple changes across the tested cuts and currents; it does not replace a field solve with a truncated interface. The complete filter results are retained in the accompanying forward-study records.

G Opt120: size, current policy and angular sampling

Between L2 and L3, changes in F, normalized by L3, reach 1.36% for the initial design and 2.71% for Opt120 across the three models on the exported mesh family; these increments are not error bounds. For Opt120 at order 45, the successive L1–L2 and L2–L3 increases in F are 3.05×10^{-8} and $2.90 \times 10^{-8}\,\mathrm{N^2\,m^2}$, so they barely shrink and do not establish an asymptotic rate. An alternate freshly constructed mesh family changes by 3.22% for Opt120 from L2 to L3. The $2.79 \times 10^{-6}\,\mathrm{N^2\,m^2}$ interface effect is about 96 times the last export-mesh increment, despite this unresolved remainder.

Current policy changes Opt120's feasibility (Fig. 8d). At L3 and order 45, the initial full-current mean is 1.4433 N m. Opt120 gives 1.3147 N m under fixed *unscaled* current, 1.2550 N m under fixed physical current, and 1.4983 N m under the reference's fixed group-1 density rule. The first two are 8.9% and 13.0% below the initial mean; the density-rule value is 3.8% above it. These policy comparisons use the same Opt120 geometry, but comparisons with the initial motor also change size: Opt120 shortens the stack from 100 to 80 mm and has radial scale $s = 1.0477$, unscaled outer diameter 130.293 mm and operating angle $-11.646°$ (Table 17). Every Opt120 torque carries the explicit size factor $s^2 L/(0.1\,\mathrm{m}) = 0.878175$. Removing both size factors from the fixed-unscaled-current mean gives 1.4971 N m, 3.7% above the initial mean. Applying the same normalization to fixed physical current gives 1.4291 N m, 1.0% below the initial mean; the density-rule value becomes 1.7061 N m, 18.2% above it. Normalizing only length to 100 mm under fixed unscaled current gives 1.6434 N m (+13.9%), retaining the radial factor $s^2 = 1.098$. These algebraic normalizations separate the size factors from the current rule; they are not reoptimized designs. Most of the saved-size deficit therefore comes from the size factor, although the normalized fixed-physical-current value remains lower.

A separate reproduction comparison uses the saved reference current at each design, rather than fixed unscaled current. At Opt120, GEODE's order-45 full-current mean is 1.4986 N m versus 1.5000 N m in the saved arrays, but its standard deviation is 3.53×10^{-4} versus 1.76×10^{-4} N m, a factor of 2.0. The half-current values are 5.53×10^{-4} and 4.37×10^{-4} N m. The largest pointwise difference across the two loaded currents is about 2.1×10^{-3} N m. With this matched current policy and zero-current values included (zero current is rule-independent and taken from the fixed-current record), the initial-to-Opt120 objective ratios are 10,170 for GEODE and 11,982 for the saved reference. The 7,631 ratio of Section E.2 instead holds unscaled current fixed. Reference refinement was run only at the initial design, so the cause of the Opt120 discrepancy remains unresolved.

Angular sampling becomes more consequential at this low ripple. Relative to the 40-angle grid, the ten-angle Opt120 objective reads about 1.05% high at order 45 and 5.90% low at order 93 on the fresh L3 mesh; the conforming value reads 17.1% low. The accompanying records retain shifted 20-angle grids and withheld currents 0.25 and 0.75. Candidate validation must check these effects at the candidate, since initial-design agreement alone would miss them. All 3,240 solve rows in the 18 successful records used for these candidate comparisons stopped at tolerance. Figure 10 shows the full and mean-subtracted waveforms.

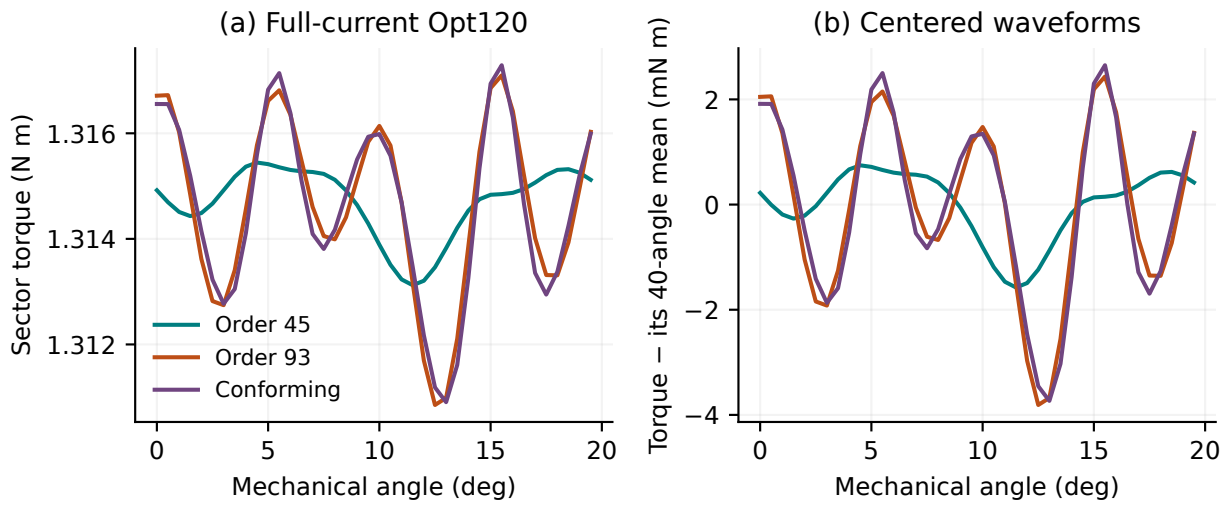


Figure 10: Opt120 full-current torque on fresh L3 meshes at 40 angles, with fixed unscaled current. Harmonic orders 45 and 93 use coupling torque; conforming coupling uses outer-band Arkkio. Left: full torque, retaining each model's mean. Right: each curve minus its own 40-angle mean. The interface change mainly alters the resolved ripple.

H Gradient verification details and derivative cost

Excitation rules

In the scalar derivative study, fixed current means fixed peak current in the unscaled geometry, so $J_g = 12 s_g I_q / |\Omega_g(X)|$ changes when a slot changes area. The reference optimizer's rule, denoted *fixed group-1 density*, sets $I_* \propto |\Omega_1(X)| s$, where s is radial scale, with the proportionality constant fixed at the initial design; the unscaled geometry is evaluated and torque is multiplied by s^2, following the reference. Both rules agree at the initial geometry but have different derivatives. Group-2 density is not held fixed because the ratio of the two winding areas can change. Rotor parameters MAG, MH and MW leave winding area unchanged and therefore give identical derivatives under the two rules. The fixed-size problem sets radial scale to one, making fixed physical and unscaled current coincide.

Taylor remainders

After removing duplicates and the analytic axial-length and radial-scale cases, 530 of 592 loaded comparisons have an anchored quadratic-remainder window spanning at least four decades in step size; 388 span at least five. The window is the contiguous run where remainder$/h^2$ stays within a factor of two of its value at $h = 10^{-5}$ m (or 10^{-2} degrees for operating angle). Smaller gradients reach round-off sooner. An anchored-window diagnostic fails for the initial-design, full-current conforming/fixed-density SW1 ripple cases at L0 and L1: at L0 the remainder changes sign near the anchor, whereas at L1 remainder$/h^2$ is still drifting there. Their widest quadratic windows span two decades. The nearly cancelling zero-current mean is a weaker check: its best relative errors reach 7.0×10^{-4}. Cogging standard-deviation errors are at most 8.9×10^{-7}. Thus the loaded agreement does not describe every zero-current derivative. Discrete derivative accuracy also does not imply mesh convergence, as the next paragraph shows.

Model dependence of verified sensitivities

The two model–estimator pairs give different ripple sensitivities even at the same design (Fig. 11). At L2 and full fixed current, the MAG ripple derivative is 1.28 N m/m for harmonic/coupling torque

versus 1.94 for conforming/Arkkio at the initial design. At the perturbed design the values are 0.75 and 1.87 N m/m. These differences persist under mesh refinement. At L3 the MAG pairs are 1.22 versus 1.87 at the initial design and 0.57 versus 1.67 N m/m at the perturbed design. In particular, the latter harmonic component changes appreciably even though the box-scaled vector changes by only 1.19% of its L2 norm; the norm itself decreases by 0.19%. Mean-torque sensitivities agree more closely in the box-scaled comparison: the Euclidean norm of the vector difference, divided by the conforming vector norm, is 0.58% and 0.66% for the mean at the initial and perturbed designs, versus 6.7% and 10.2% for ripple. The norm weights the largest components most heavily: the initial SW2 and SW3 mean components differ by 6.5% and 16.5% relative to conforming values. At L3, the ripple gaps remain 6.58% and 10.25%, while each model changes by 0.46–1.19% from L2 to L3. The mean gaps fall to 0.16% and 0.26%; the conforming model itself changes by 0.52–0.54%. The L2 mean gap is therefore comparable to spatial refinement, whereas the ripple gap persists beyond the measured refinement increments. Because the estimator changes with the model, this experiment measures their combined effect; it does not isolate an interface-only derivative error or establish which gradient is closer to the physical machine.

Discrete derivative accuracy also does not imply mesh convergence. In the harmonic model, SW3 mean-torque sensitivities change sign between coarse levels and their increments shrink with refinement; ripple gradients converge more slowly than most mean gradients. The conforming/Arkkio SW2 mean sensitivity is non-monotone across L0–L3 (-46.5, -41.3, -44.4, -47.2 N m/m), whereas harmonic/coupling torque approaches -47.1 from -49.8 N m/m. Fresh-remesh checks of MW, SW1 and SW2 are limited at small steps by changes in connectivity. The L2 MW mean-derivative fit has relative gap 2.6×10^{-6}, but individual remeshed steps at $h \geq 10^{-4}$ m scatter by about 6×10^{-4}. The largest L2 full-current remeshed-versus-moved fitted discrepancy is 3.8% for the SW1 ripple derivative. These checks support the discrete chain rule while leaving refinement of candidate designs necessary.

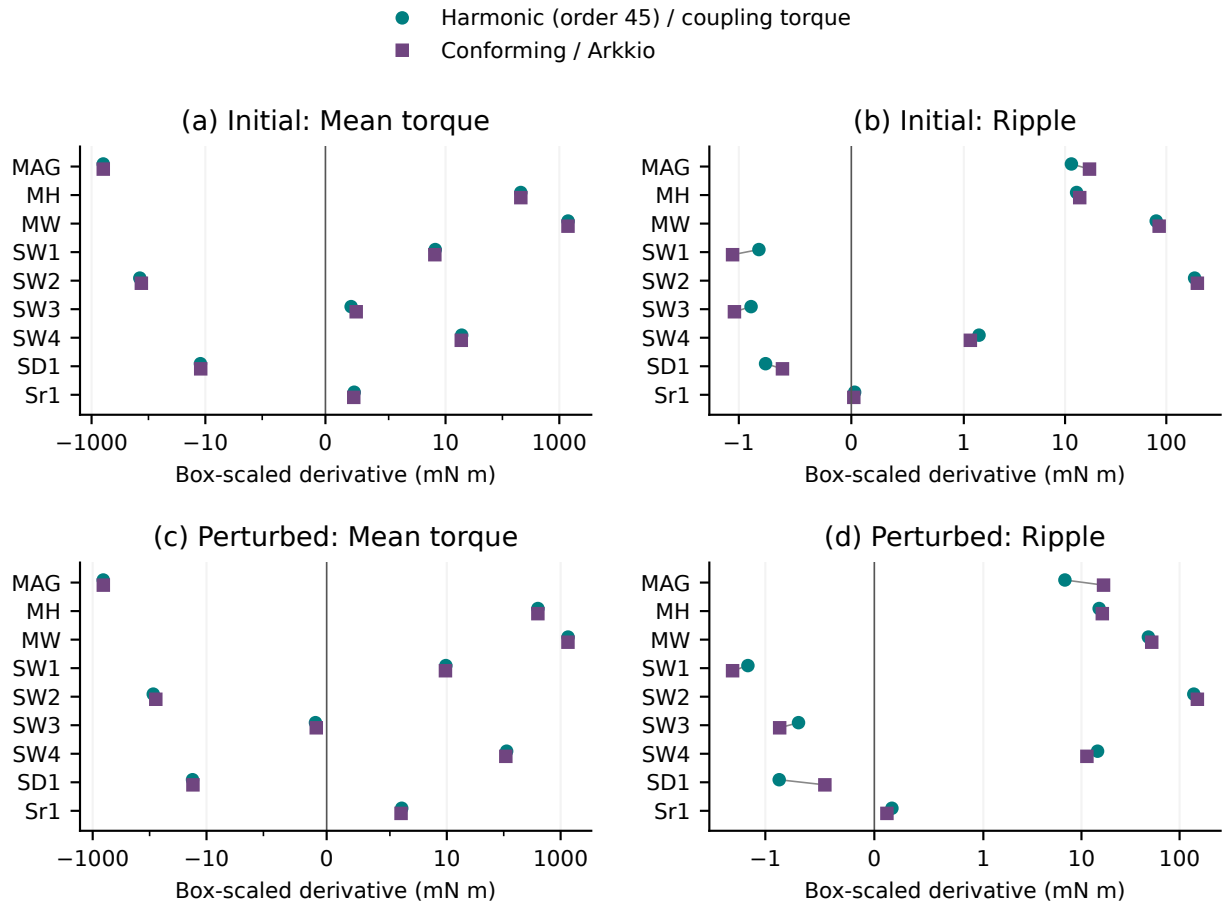


Figure 11: Full fixed-current sensitivities at L2, scaled by each geometric parameter's reference box width w_k. Top: initial design; bottom: perturbed design. L3 MAG values and refinement changes are given in the text. Left: mean torque; right: standard deviation. The plotted quantity $w_k \partial q / \partial p_k$ is a local sensitivity, not a finite change across the box. Coupling and torque estimator both differ between the two series. Gray segments join the two estimates for each parameter. Horizontal axes are symmetric-log, linear within ± 1 mN m. All nine widths, in mm, are 9, 10, 15, 4, 3, 1, 18, 70 and 1.5 in the displayed order.

The recorded timings use an Apple M3 Ultra with 28 cores, default solver threading, and concurrent host load averaging 14–24. The effective thread count was not pinned. A forward evaluation covers ten angles, starts from zero, and warm-starts successive angles. The full gradient includes that forward, ten torque adjoints, mesh derivatives and contractions for all 12 parameters and both aggregate objectives. Central differences use 24 forward sweeps, each warm-started from the base states, at step $h = 10^{-6}$ in each parameter's stored unit (m, degrees or dimensionless scale), plus the shared base forward for a common starting point. Across L0–L2, both couplings and tolerances 10^{-8} and 10^{-14}, the full gradient costs 1.13–1.20 forwards, whereas central differences cost 8.2–10.5 including the shared base solve (Fig. 3d and Table 16). Shape-map setup is a separate one-time cost of these timings; the optimization study charges whole-run wall time, so it is included there. These derivative timings concern one loaded host and are a separate workload from the 1,640 forward validation solves of Section E.

Table 16: Recorded harmonic-model cost at relative tolerance 10^{-14}, initial design and full current. Times are seconds for ten-angle sweeps; FD includes the shared base sweep, and setup is a one-time shape-map cost.

Mesh	Forward	Forward + gradient	Base + FD	Setup
L0	0.664	0.748	6.997	0.011
L1	3.174	3.658	32.811	0.041
L2	12.702	14.644	129.390	0.222

I Linear slotless screening control

The stored slotless model has 24 distinct angles. Fitting $T(\phi) = c_0 + c_1 \cos\phi + c_2 \sin\phi$, with electrical angle $\phi = p\theta_r$, mechanical rotor angle θ_r and pole-pair count p, on alternate angles and evaluating on the other twelve gives normalized held-out RMSE 3.08×10^{-10} and maximum error 6.82×10^{-10}, using the recorded analytic-amplitude normalization. This is a locked-rotor angle sweep, not synchronous operating ripple. For a fixed linear field operator and sinusoidal sources, the field lies in a small basis of source responses; a quadratic torque functional can contain only the corresponding finite set of harmonics. The particular model needs just the three fitted coefficients. This is an angular interpolation test at one geometry, not a shape-gradient test or evidence of generalization across designs.

J Design points for derivative checks

Table 17 identifies the scalar derivative-study coordinates and the saved Opt120 coordinates. Values are rounded for display; the accompanying records retain full precision. The operating angle is stored in degrees and converted to radians in the current equation. These coordinates describe the scalar design space; the saved reference optimum additionally contains surface-control offsets.

Table 17: Initial and interior perturbed design points for derivative checks, alongside Opt120 for the forward validation. Lengths refer to the unscaled geometry; surface-control offsets are retained in the data.

Parameter	Meaning	Unit	Initial	Perturbed	Opt120
MAG	Pole-axis iron depth above magnet	mm	7.0000	6.5506	6.1011
MH	Radial magnet height	mm	7.0000	4.8722	2.7444
MW	Magnet width	mm	19.0000	19.8670	20.7340
SW1	Stator tooth width	mm	4.0000	4.0309	4.0619
SW2	Slot-opening width	mm	2.3000	1.6500	1.0001
SW3	Slot-opening depth	mm	1.0000	0.7502	0.5004
SW4	Back-iron depth on slot axis	mm	8.2500	6.8418	5.4336
SD1	Stator outer diameter	mm	135.0000	132.6466	130.2931
Sr1	Slot-bottom fillet radius	mm	1.0000	1.4999	1.9998
LENGTH	Axial length	mm	100.0000	90.0003	80.0005
ϕ_0	Electrical current offset	°	0.0000	−5.8231	−11.6463
ScaleR	Radial similarity scale	1	1.0000	1.0239	1.0477

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